Topology Optimization
Lecture 10

Creating geometrically robust designs for highly sensitive problems using topology optimization

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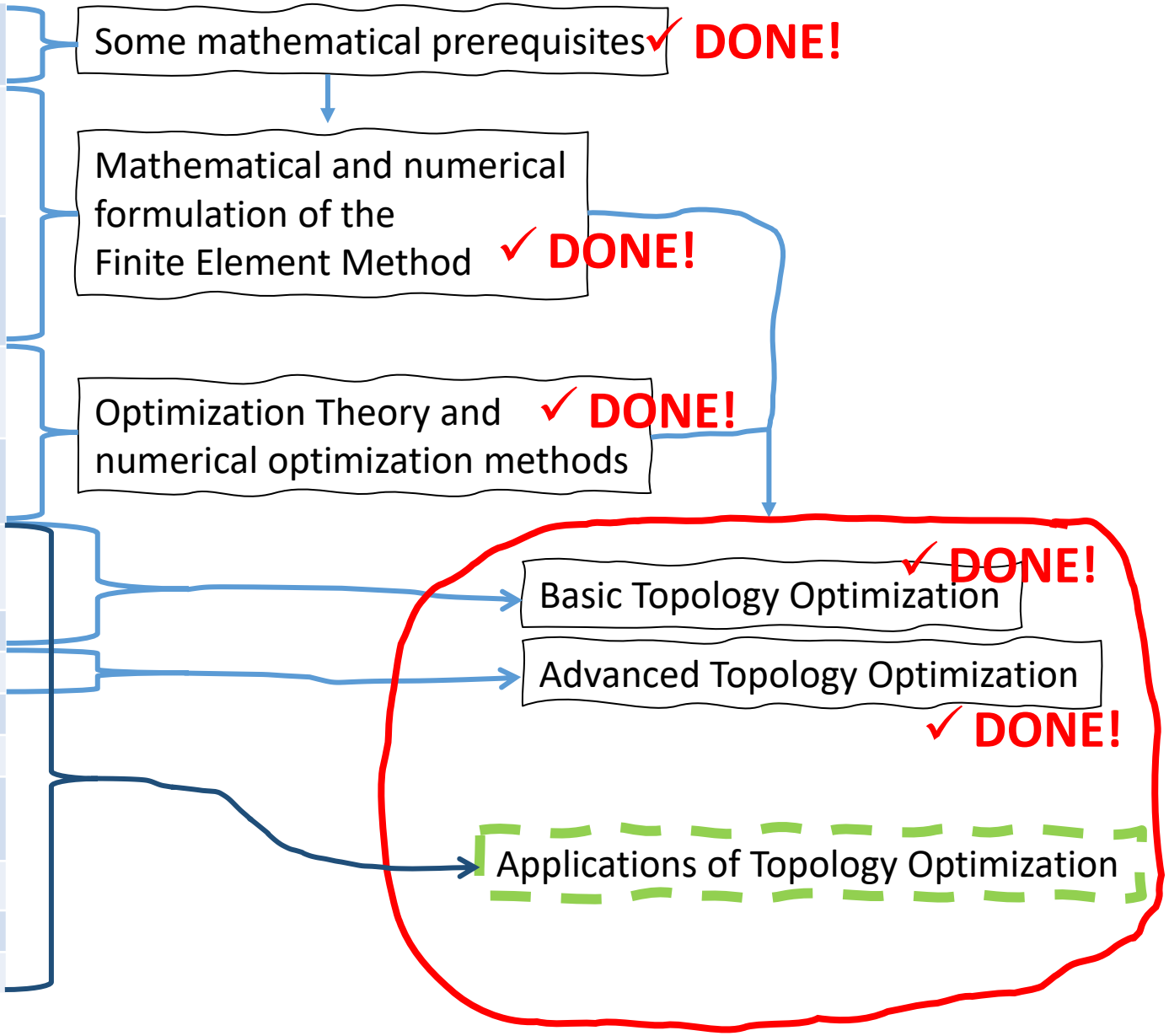
Course overview

No.	Topics
1	Introduction Function and vector spaces, Interpolation, L^2 projection
2	The Finite Element Method for PDEs 1: Poisson's Equation, The Finite Element Method, Properties of the Finite Element Method
3	The Finite Element Method for PDEs 2: Index notation, Linear Elasticity, Non-linear Poisson's equation, Non-linear Elasticity
4	Constrained Optimization 1: Optimality, KKT Theorem, Numerical Optimization Algorithms
5	Constrained Optimization 2: Method of Moving Asymptotes, MPI and PETSc
6	Introduction to Topology Optimization: The idea, Filtering and Projection, Python implementation
7	Gradients
8	Robust topology optimization, More on MPI
9	Wave equations
10	Acoustic waves and more on Robust Optimization
11	Penalty and Augmented Lagrangian Methods Stress-constraints in Topology Optimization 1
12	Stress-constraints in Topology Optimization 2
13	Light concentration using nanoparticle design
14	Heat conduction



Recap: Sound Waves (in homogeneous media)

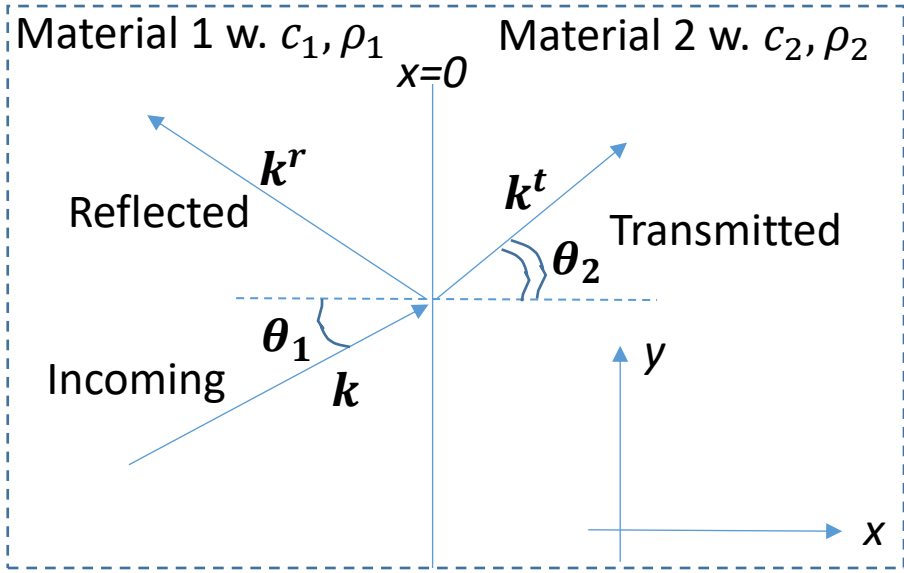
- The speed of sound and density enter directly in the wave equation and the velocity equation:

$$\frac{\partial^2 p}{\partial t^2} - c^2 \nabla^2 p = 0$$

$$\rho_0 \frac{\partial \boldsymbol{u}}{\partial t} + \nabla p = 0$$

- The speed of sound varies greatly from gas, to liquid, to solid:
  - In air is ~340 m/s=1224 km/h.
  - In water ~1500 m/s=5400 km/h.
  - In aluminum ~6300 m/s=22680 km/h.
- People also sometimes introduce the following quantities in wave problems:
  - $n = \frac{c_0}{c}$, the refractive index as the ratio between a reference speed c_0 and the wave speed in the medium, c.
  - $Z = c\rho_0$, the impedance of the medium.
  - $K = \rho_0 c^2$, the adiabatic bulk modulus.

Recap: Reflection and Refraction at Interfaces



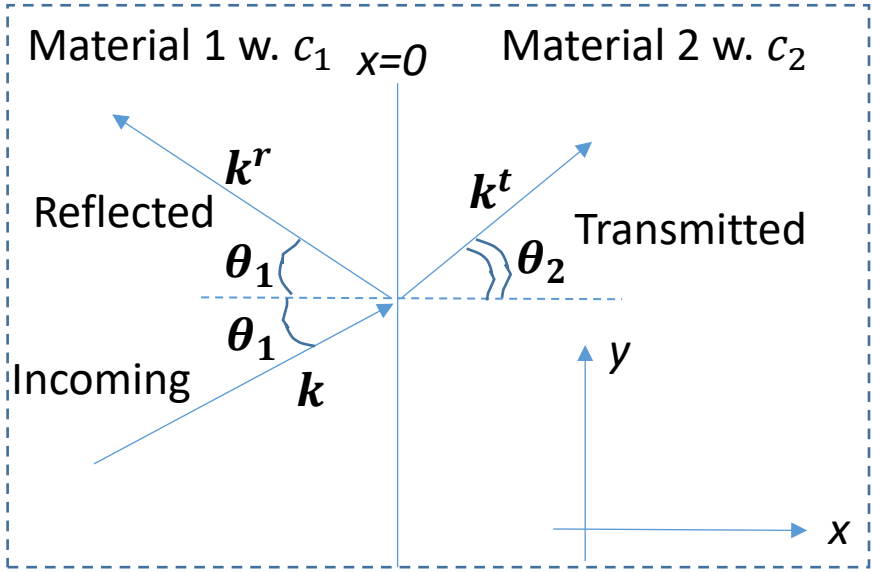
- We take the incoming wave to have an angle θ_1 with the interface normal.
- The interface is at *x*=0.
- For x<0 we write $p = Ie^{i(\omega t - \boldsymbol{k}\cdot\boldsymbol{r})} + Re^{i(\omega t - \boldsymbol{k^r}\cdot\boldsymbol{r})}$ as there is an incoming wave with amplitude I and a reflected wave with amplitude R.
- For x>0 we write $p = Te^{i(\omega t - \boldsymbol{k^t}\cdot\boldsymbol{r})}$ as there is a transmitted wave with amplitude T.
- We write $\boldsymbol{k} = \begin{pmatrix} k_1 \cos\theta_1 \\ k_1 \sin\theta_1 \end{pmatrix}$, $\boldsymbol{k^r} = \begin{pmatrix} k_x^r \\ k_y^r \end{pmatrix}$, $\boldsymbol{k^t} = \begin{pmatrix} k_x^t \\ k_y^t \end{pmatrix}$.
- Demanding the pressure to be equal at the interface, we get
$$Ie^{i(\omega t - \boldsymbol{k}\cdot\boldsymbol{r})}\Big|_{x=0} + Re^{i(\omega t - \boldsymbol{k^r}\cdot\boldsymbol{r})}\Big|_{x=0} = Te^{i(\omega t - \boldsymbol{k^t}\cdot\boldsymbol{r})}\Big|_{x=0} \quad \Rightarrow \quad Ie^{-ik_1 \sin\theta_1 y} + Re^{-ik_y^r y} = Te^{-ik_y^t y} \quad \Rightarrow \quad k_1 \sin\theta_1 = k_y^r = k_y^t$$
since it must be true for all values of y.
- Since $c = \frac{\omega}{k}$ for each medium then $|\boldsymbol{k}| = k_1 = \frac{\omega}{c_1}$ and $|\boldsymbol{k^r}| = \frac{\omega}{c_1} = k_1$ and $|\boldsymbol{k^t}| = \frac{\omega}{c_2} = k_2$.
- For $\boldsymbol{k^r} = \begin{pmatrix} k_x^r \\ k_1 \sin\theta_1 \end{pmatrix}$ we get $k_x^r = -k_1 \cos\theta_1$ and thus $\boldsymbol{k^r} = \begin{pmatrix} -k_1 \cos\theta_1 \\ k_1 \sin\theta_1 \end{pmatrix}$ - "the incident angle is equal to the reflected angle".
- Writing $\boldsymbol{k^t} = \begin{pmatrix} k_x^t \\ k_y^t \end{pmatrix} = \begin{pmatrix} k_2 \cos\theta_2 \\ k_2 \sin\theta_2 \end{pmatrix} = \begin{pmatrix} k_2 \cos\theta_2 \\ k_1 \sin\theta_1 \end{pmatrix}$ we get Snell's law:
$$k_2 \sin\theta_2 = k_1 \sin\theta_1$$
- Useful to determine θ_2 from θ_1 and the material properties.

Recap: Reflection and Refraction at Interfaces

- Snell's law: $k_2 \sin\theta_2 = k_1 \sin\theta_1$ we got from matching the pressure at the material interface.
- The coefficients *R* and *T* can be found by also matching the velocities at the interface using $\rho \frac{\partial \boldsymbol{u}}{\partial t} + \nabla p = 0$.
- The results (derive yourself ☺) can be written as

$$r(\theta_1) = \frac{R}{I} = \frac{Z_2 \cos\theta_1 - Z_1 \cos\theta_2}{Z_2 \cos\theta_1 + Z_1 \cos\theta_2}$$

$$t(\theta_1) = \frac{T}{I} = \frac{2Z_2 \cos\theta_1}{Z_2 \cos\theta_1 + Z_1 \cos\theta_2}$$



- $Z_1 = \rho_1 c_1$ and $Z_2 = \rho_2 c_2$ are the impedances of the two media.
- In optics, these are known as Fresnel coefficients for TE polarized light.
- For normal incidence, $\theta_1 = 0$ and $r(0) = \frac{Z_2 - Z_1}{Z_2 + Z_1}$.
- If $Z_2 \gg Z_1$ or $Z_1 \gg Z_2$ we get $|r| \approx 1$ and the amplitude of the reflected wave is equal to the amplitude of the incident wave.
- If $Z_2 \approx Z_1$, $|r| \approx 0$ and there is almost no reflected wave. This is named impedance matching.

Sound Waves in Inhomogeneous Media

(for self study)

- The fundamental equations of fluid dynamics can be written as

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0$$
$$\frac{\partial \rho \boldsymbol{u}}{\partial t} + \nabla \cdot (\rho \boldsymbol{u} \otimes \boldsymbol{u} + \tau) = -\nabla p + \rho \boldsymbol{g}$$
$$\rho T \frac{\partial s}{\partial t} + \rho T \boldsymbol{u} \cdot \nabla s = \tau : \nabla \boldsymbol{u} - \nabla \cdot \boldsymbol{q}$$

- Considering small perturbations (‘ quantities) to a fluid at rest, without gravity and heat flux, we write

$$\rho = \rho_0 + \rho', \qquad \boldsymbol{u} = \boldsymbol{u}_0 + \boldsymbol{u}', \qquad p = p_0 + p', \qquad T = T_0 + T', \qquad s = s_0 + s'$$

with $\boldsymbol{u}_0 = \boldsymbol{g} = \boldsymbol{q} = 0$.

- To be more general than previous, we assume p_0 is constant, no 0-quantity change with time, viscous effects are not important and neglect higher orders of the ‘ quantities. The equations reduce to

$\frac{\partial(\rho_0+\rho')}{\partial t} + \nabla \cdot \left((\rho_0 + \rho')\boldsymbol{u}'\right) = 0 \;\Rightarrow\; \frac{\partial \rho'}{\partial t} + \nabla \cdot (\rho_0 \boldsymbol{u}') = 0$ (no change)

$\frac{\partial(\rho_0+\rho')\boldsymbol{u}'}{\partial t} + \nabla \cdot \left((\rho_0 + \rho')\boldsymbol{u}' \otimes \boldsymbol{u}' + \tau(\boldsymbol{u}')\right) = -\nabla(p_0 + p') \;\Rightarrow\; \rho_0 \frac{\partial \boldsymbol{u}'}{\partial t} + \nabla p' = 0$ (no change)

$(\rho_0 + \rho')(T_0 + T')\frac{\partial(s_0+s')}{\partial t} + (\rho_0 + \rho')(T_0 + T')\boldsymbol{u}' \cdot \nabla(s_0 + s') = 0 \Rightarrow \rho_0 T_0 \frac{\partial s'}{\partial t} + \rho_0 T_0 \boldsymbol{u}' \cdot \nabla s_0 = 0 \;\Rightarrow\; \frac{\partial s'}{\partial t} + \boldsymbol{u}' \cdot \nabla s_0 = 0$ (new)

- This new equation can also be stated as

$$\frac{\partial s'}{\partial t} + \boldsymbol{u}' \cdot \nabla s_0 = \frac{Ds}{Dt} = 0$$

- $\frac{D}{Dt}$ is the co-called material derivative.

Sound Waves in Inhomogeneous Media

(for self study)

- To close the system, an equation of state expressing the pressure is needed:

$$p(\rho, s) \Rightarrow dp = \left(\frac{\partial p}{\partial \rho}\right)\Big|_s d\rho + \left(\frac{\partial p}{\partial s}\right)\Big|_\rho ds = c^2 d\rho + \left(\frac{\partial p}{\partial s}\right)\Big|_\rho ds$$

- $c(\rho, s)$ is the isentropic speed of sound.
- Writing the material derivative of the pressure and using $\frac{Ds}{Dt} = 0$ yield

$$\frac{Dp}{Dt} = c^2 \frac{D\rho}{Dt} + \left(\frac{\partial p}{\partial s}\right)\Big|_\rho \frac{Ds}{Dt} = c^2 \frac{D\rho}{Dt} \quad \Rightarrow \quad \frac{\partial p}{\partial t} + \boldsymbol{u} \cdot \nabla p = c^2 \left(\frac{\partial \rho}{\partial t} + \boldsymbol{u} \cdot \nabla \rho\right)$$

- Using the assumptions from last slide and keeping only linear terms in ‘ quantities, we get

$$\frac{\partial (p_0 + p')}{\partial t} + \boldsymbol{u}' \cdot \nabla (p_0 + p') = c^2 \left(\frac{\partial (\rho_0 + \rho')}{\partial t} + \boldsymbol{u}' \cdot \nabla (\rho_0 + \rho')\right) \Rightarrow$$

$$\frac{\partial p'}{\partial t} + \boldsymbol{u}' \cdot \nabla p_0 = \frac{\partial p'}{\partial t} = c^2 \left(\frac{\partial \rho'}{\partial t} + \boldsymbol{u}' \cdot \nabla \rho_0\right)$$

Sound Waves in Inhomogeneous Media

(for self study)

- The equations are now

$$\frac{\partial \rho'}{\partial t} + \nabla \cdot (\rho_0 \boldsymbol{u}') = 0 \qquad (*)$$

$$\rho_0 \frac{\partial \boldsymbol{u}'}{\partial t} + \nabla p' = 0 \qquad (**)$$

$$\frac{\partial p'}{\partial t} = c^2 \left(\frac{\partial \rho'}{\partial t} + \boldsymbol{u}' \cdot \nabla \rho_0 \right) \qquad (***)$$

- Inserting $(*)$ into $(***)$ we get

$$\frac{\partial p'}{\partial t} = c^2 \left(\frac{\partial \rho'}{\partial t} + \boldsymbol{u}' \cdot \nabla \rho_0 \right) = c^2 (-\nabla \cdot (\rho_0 \boldsymbol{u}') + \boldsymbol{u}' \cdot \nabla \rho_0) = c^2 (-\rho_0 \nabla \cdot \boldsymbol{u}' - \boldsymbol{u}' \cdot \nabla \rho_0 + \boldsymbol{u}' \cdot \nabla \rho_0) = -c^2 \rho_0 \nabla \cdot \boldsymbol{u}'$$

- Taking the time-derivative and using $(**)$, result in

$$\frac{\partial^2 p'}{\partial t^2} = -c^2 \rho_0 \nabla \cdot \frac{\partial \boldsymbol{u}'}{\partial t} = -c^2 \rho_0 \nabla \cdot \left(-\frac{1}{\rho_0} \nabla p' \right) \quad \Rightarrow \quad \frac{1}{\rho_0 c^2} \frac{\partial^2 p'}{\partial t^2} - \nabla \cdot \left(\frac{1}{\rho_0} \nabla p' \right) = 0$$

- Dropping the ‘, using $\rho_0 c^2 = K$ and the time-harmonic ansatz $\left(p(\boldsymbol{r}, t) = p(\boldsymbol{r}) e^{i\omega t} \right)$, the equation become:

$$\nabla \cdot (\rho_0^{-1} \nabla p) + \omega^2 K^{-1} p = 0$$

- These forms of the acoustic wave equations are appropriate when the “background density”, ρ_0, change with position, i.e., when multiple materials are present.

Sound pressure in frequency domain

Imaginary numbers:
$i^2 = -1$
$z = x + iy$
$\mathrm{Re}(z) = x$
$\mathrm{Im}(z) = y$
Euler's Formula:
$e^{ix} = \cos(x) + i\sin(x)$

- To get into the frequency domain, the time-harmonic ansatz
$$p(\boldsymbol{r}, t) = \mathrm{Re}\left(p(\boldsymbol{r})e^{i\omega t}\right)$$
was used in last lecture. Often without the Re function (real part). Re should then be taken at the end to get the physical field.

- Let $p(\boldsymbol{r})$ be a complex field, and write
$$p(\boldsymbol{r}) = p_r(\boldsymbol{r}) + ip_i(\boldsymbol{r}) = |p(\boldsymbol{r})|e^{i\phi(\boldsymbol{r})}$$
$p_r(\boldsymbol{r}), p_i(\boldsymbol{r}), |p(\boldsymbol{r})|, \phi(\boldsymbol{r})$ are all real fields.

- Then
$$p(\boldsymbol{r}, t) = \mathrm{Re}\left(p(\boldsymbol{r})e^{i\omega t}\right) = \mathrm{Re}\left(|p(\boldsymbol{r})|e^{i\phi(\boldsymbol{r})}e^{i\omega t}\right) = |p(\boldsymbol{r})|\mathrm{Re}\left(e^{i\left(\omega t + \phi(\boldsymbol{r})\right)}\right)$$
$$= |p(\boldsymbol{r})|\mathrm{Re}\left(\cos\left(\omega t + \phi(\boldsymbol{r})\right) + i\sin\left(\omega t + \phi(\boldsymbol{r})\right)\right) = |p(\boldsymbol{r})|\cos\left(\omega t + \phi(\boldsymbol{r})\right)$$

- The amplitude of the sound pressure is thus the norm of the complex sound pressure:
$$|p(\boldsymbol{r})| = \sqrt{\left(p_r(\boldsymbol{r})\right)^2 + \left(p_i(\boldsymbol{r})\right)^2}$$

- $|p(\boldsymbol{r})|$ tells us how 'loud' the sound is, and it is therefore often the quantity of interest.

- The following handy relation is often used
$$|p(\boldsymbol{r})|^2 = p(\boldsymbol{r})p^*(\boldsymbol{r})$$

- $p^*(\boldsymbol{r}) \equiv p_r(\boldsymbol{r}) - ip_i(\boldsymbol{r}) = |p(\boldsymbol{r})|e^{-i\phi(\boldsymbol{r})}$ is the complex conjugate.

- $L_p = 10\log_{10}\left(\frac{|p|^2}{p_{ref}^2}\right)$, with $p_{ref} = 20\mu\mathrm{Pa}$, is named the "Sound Pressure Level", https://en.wikipedia.org/wiki/Sound_pressure#Sound_pressure_level

The main points of the paper

- Single frequency wave propagation problems can be very sensitive to design variations, especially for high frequencies.
- The standard filtering and projection approach can result in geometries which are very sensitive towards design variations and thus almost impossible to manufacture.
- Robust topology optimization does not work satisfactorily either, so a new robust formulation is proposed.
- The new robust formulation is named the double filter approach.
- The approach works like this:
  - A set of K thresholds is defined: $\eta = \{\eta^1, \eta^2, \dots, \eta^K\}$. $K = 3$ is often used.
  - The design variables are filtered and projected with threshold $\eta_1 = \min \eta$ and strength β_1.
  - The filtered and projected design variables are filtered and projected again with strength β_2 using the thresholds in the set η to create K different "eroded"/"dilated" designs.
  - The double filtering ensures a more consistent topology between the eroded, nominal and dilated designs.
- The new double filtered robust approach is shown to be able to create geometrically robust designs for very sensitive wave propagation problems (usual with a single frequency).
- The approach also works well for "normal" problems.

The model in the paper

- The time-harmonic wave equation is solved for a single frequency to obtain the pressure waves.
- The equation considered in the paper is $\nabla \cdot (\rho^{-1}\nabla p) + \omega^2 K^{-1} p = 0$ with velocity equation $i\omega \boldsymbol{u} + \rho^{-1}\nabla p = 0$.
- The model is a cavity, and $\boldsymbol{u} = 0$ are used on all sides to model hard walls.
- Most of the domain is filled with air.
- A source is located to the left and a target domain to the right.
- The top of the domain (from 8 m to 9 m) is the design domain, where the optimizer can place either air or aluminum.
- The sound pressure, p, is minimized in the optimization domain, Ω_{OD}.
- Frequencies of 34.36 Hz, 51.32 Hz ,69.42 Hz and 206.3 Hz are considered.
- SIMP interpolation is NOT used, instead linear interpolation is used in the inverse material density and bulk modulus:

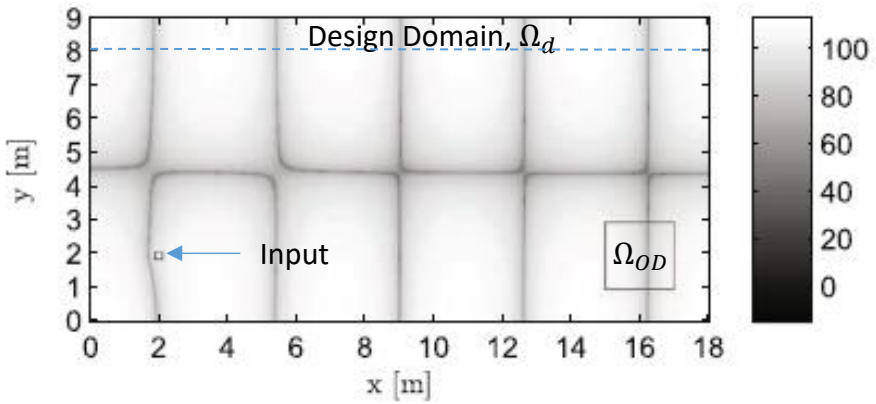
$$\rho^{-1}(\xi) = \frac{1}{\rho(\xi)} = \frac{1}{\rho_{air}} + \xi\left(\frac{1}{\rho_{Al}} - \frac{1}{\rho_{air}}\right), \qquad K^{-1}(\xi) = \frac{1}{K(\xi)} = \frac{1}{K_{air}} + \xi\left(\frac{1}{K_{Al}} - \frac{1}{K_{air}}\right)$$

ξ denotes the filtered and projected design variables.

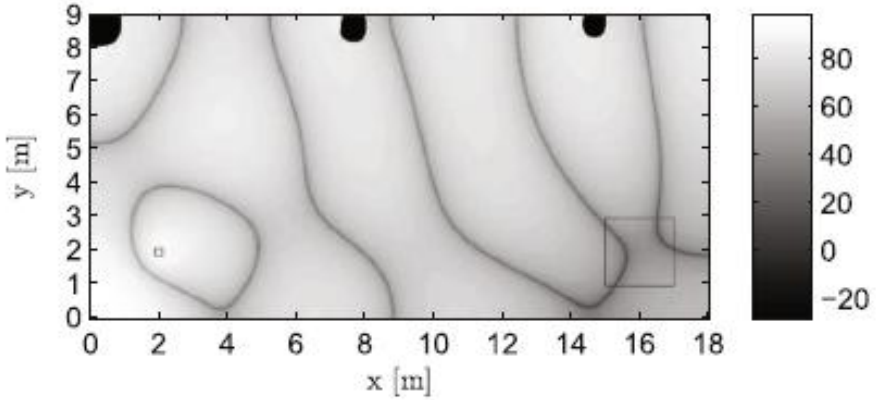
- The optimization problem considered is

$$\min_{\xi} \frac{\int_{\Omega_{OD}} |p(\xi)|^2 d\Omega}{\int_{\Omega_{OD}} d\Omega}, \qquad \text{s.t. } 0 \le \xi \le 1$$

- The optimization problem is solved using MMA.



(i) $L_{\hat{p}}$ in empty model domain.



(ii) $L_{\hat{p}}$ in model domain where optimized design has been introduced.

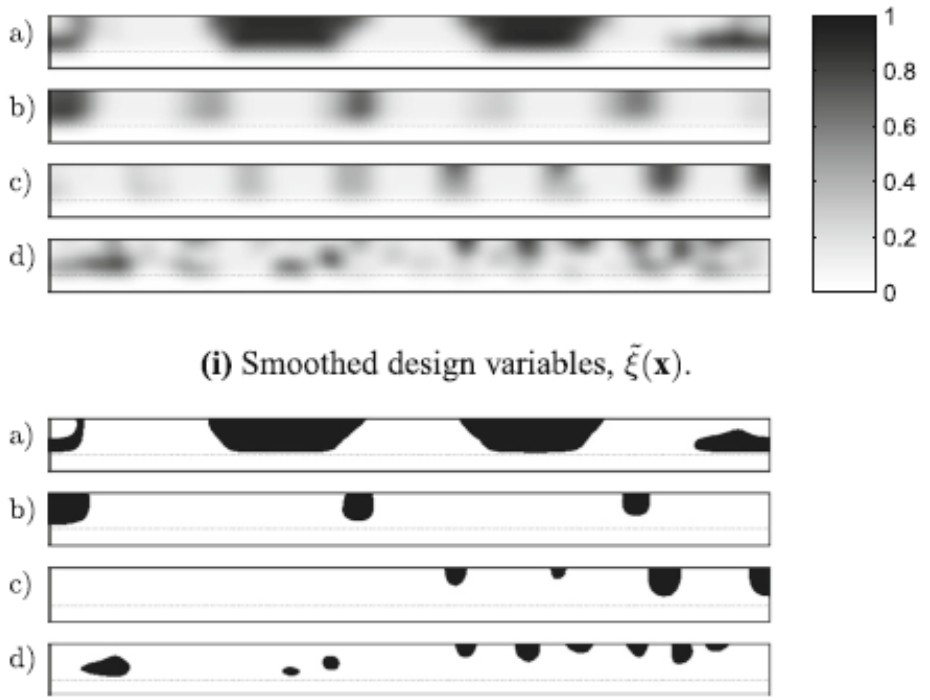
Fig. 2 Pressure fields measured using $L_{\hat{p}}$ at the excitation frequency $f = 51.32$ Hz. The acoustic source and Ω_{OP} are outlined using thin black lines

The standard approach

- The standard approach is using a single filtering and projection.
- Filtering is done using a convolution-based filter.
- Filtered variables are denoted with an over-tilde ~.
- Projection is denoted with an overbar – and is done with the well-known formula

$$\bar{\xi}_i = \frac{\tanh(\beta\eta) + \tanh(\beta(\xi_i - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}, \tag{23}$$

- The standard approach uses $\eta = 0.5$.
- The projection strength, β, is raised by a factor of 1.2 and not the "usual" factor of 2.
- The approach is shown to produce design with a lot of intermediate values in the design variables.
- The intermediate values are projected to 0 or 1, and the filtered and projected densities, $\bar{\tilde{\xi}}$, are still close to 0 or 1 at the end of the optimization.



(i) Smoothed design variables, $\tilde{\xi}(\mathbf{x})$.

(ii) Final designs. (Physical design variables, $\bar{\tilde{\xi}}(\mathbf{x})$ projected at $\eta = 0.5$).

Fig. 3 **(i)** Smoothed, $\tilde{\xi}$, and **(ii)** physical, $\bar{\tilde{\xi}}$, design variables for four excitation frequencies, **a)** $f = 34.36$ Hz, **b)** $f = 51.32$ Hz, **c)** $f = 69.42$ Hz, **d)** $f = 206.3$ Hz. The standard approach has been used for the optimization. The dashed line denote the edge of $\mathbf{\Omega}_d$

The standard approach

- The standard approach produce designs with good performance.
- $L_p = 10\log_{10}\left(\frac{|p|^2}{p_{ref}^2}\right)$, with $p_{ref} = 20\mu$Pa, is the "Sound Pressure Level".
- It is stated that the designs are very sensitive to changes in the geometry, which means they are impossible to produce while keeping the high performance.
- The images on the right, from a similar optimization, show that very small changes in the design can lead to drastic changes in the sound pressure.
- A robust approach is thus preferred.

Table 2 Average sound pressure level in Ω_{OP} at four different excitation frequencies, for the cases: Empty design domain, $\langle L_{\hat{p}}\rangle_{\Omega_{OP_{ED}}}$. Design domain filled with material; $\langle L_{\hat{p}}\rangle_{\Omega_{OP_{FD}}}$. Optimized design introduced, $\langle L_{\hat{p}}\rangle_{\Omega_{OP_{OD}}}$

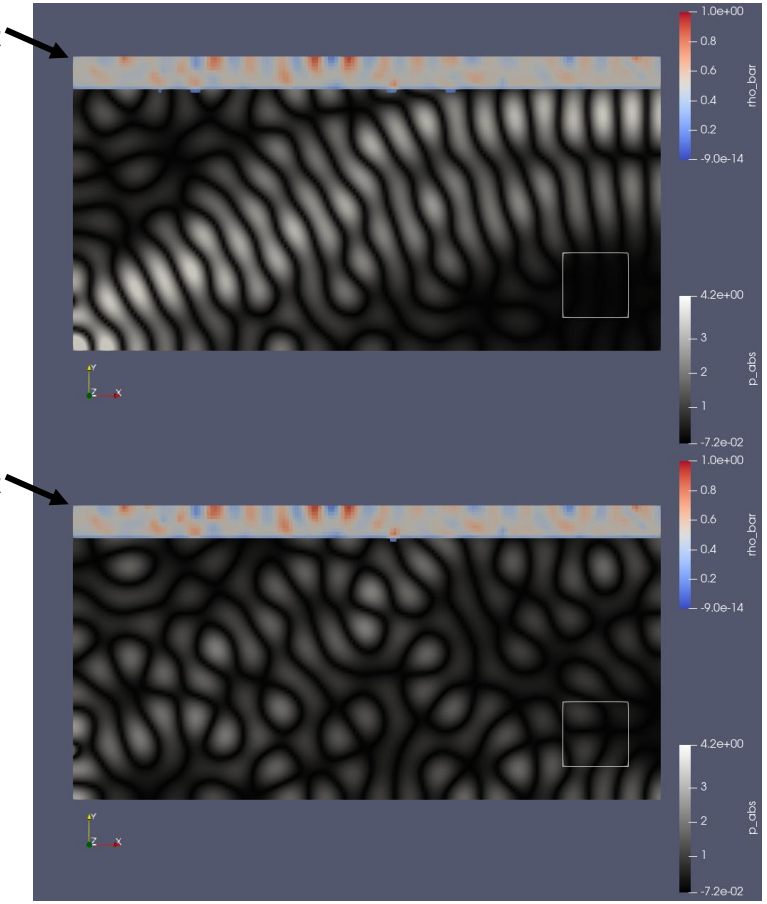
f [Hz]	$\langle L_{\hat{p}}\rangle_{\Omega_{OP_{ED}}}$ [dB]	$\langle L_{\hat{p}}\rangle_{\Omega_{OP_{FD}}}$ [dB]	$\langle L_{\hat{p}}\rangle_{\Omega_{OP_{OD}}}$ [dB]
34.36	≈ 114	≈ 80.5	≈ 64.4
51.32	≈ 99.0	≈ 59.0	≈ 38.8
69.42	≈ 127	≈ 91.8	≈ 46.1
206.3	≈ 120	≈ 116	≈ 64.4

Design Domain, Ω_d

MMA
Iteration 10

Design Domain, Ω_d

MMA
Iteration 11



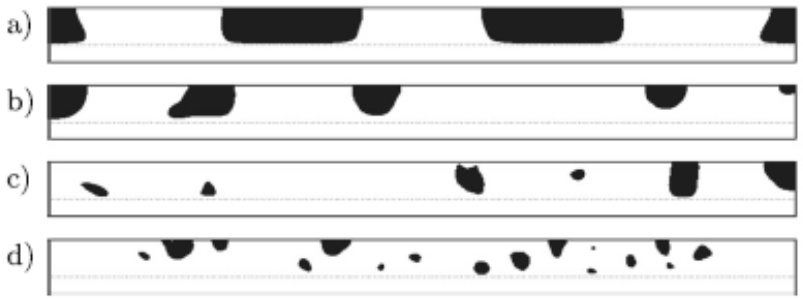
The robust approach

- The robust approach, as you implemented in the compliant mechanism exercise, is also considered.
- Three designs are considered in the min-max problem, the eroded ($\eta = 0.7$), the nominal ($\eta = 0.5$) and the dilated design ($\eta = 0.3$).
- The optimization problem considered for the robust approach is a min-max problem:

$$\min_{\xi} \max_{k} \frac{\int_{\Omega_{OD}} \left| p\left(\bar{\bar{\xi}}_k\right) \right|^2 d\Omega}{\int_{\Omega_{OD}} d\Omega}, \qquad \text{s.t.} \quad 0 \leq \xi \leq 1$$

- k denotes the design realizations, e.g.,
  - $k = 0$ is the nominal design.
  - $k = 1$ is the dilated design.
  - $k = 2$ is the eroded design.
- Only the low frequency design is similar to the design found with the standard approach.
- The filtered variables are still seen to have many intermediate valued regions (grey regions).

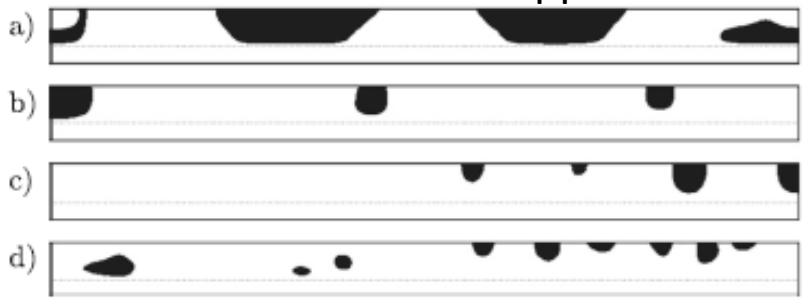
(i) Smoothed design variables



(ii) Final designs, projected at $\eta = 0.5$.

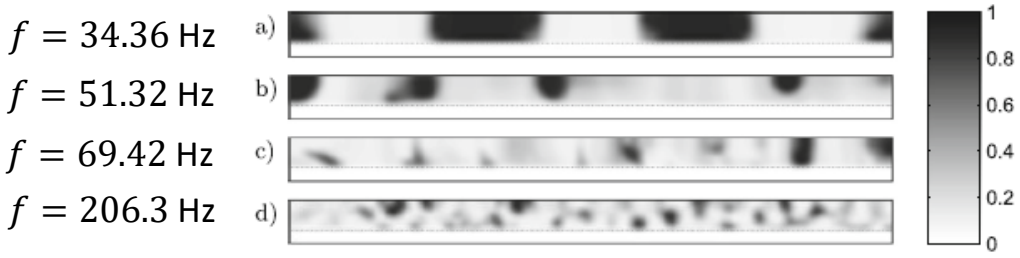
Fig. 5 **(i)** Smoothed and **(ii)** projected design variables for the four excitation frequencies, **a)** $f = 34.36$ Hz, **b)** $f = 51.32$ Hz, **c)** $f = 69.42$ Hz, **d)** $f = 206.3$ Hz, obtained using the robust approach given in (27) with a smoothing and projection step

From the standard approach

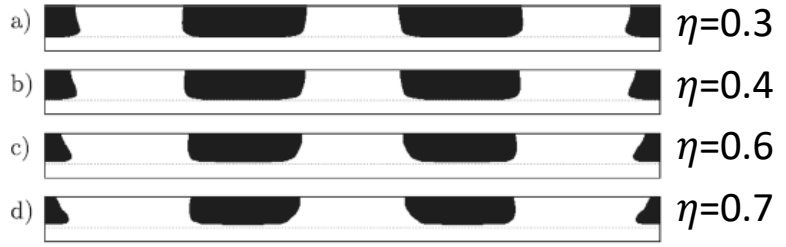


The robust approach

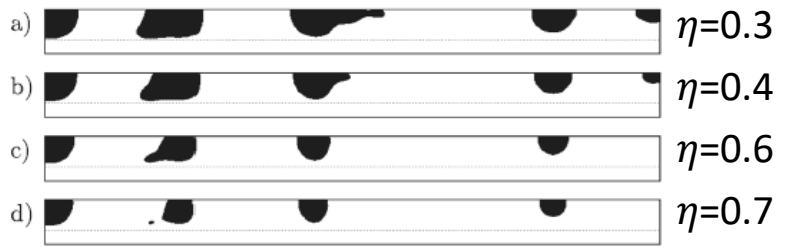
- The intermediate values makes the designs with different projection thresholds (η) have different topologies.
- It was assumed in the robust approach that the overall topology of the design was not changing when changing η.
- Only the low frequency design yield the same topology for different projection levels, and the robust approach is thus expected to work well at low frequencies.
- Many filtered design variables have values close to the projection threshold, resulting in a difference in the overall topology.
- For the high frequency cases, a new robust approach is invented in the paper.
- The new approach is named “The double filter approach”.



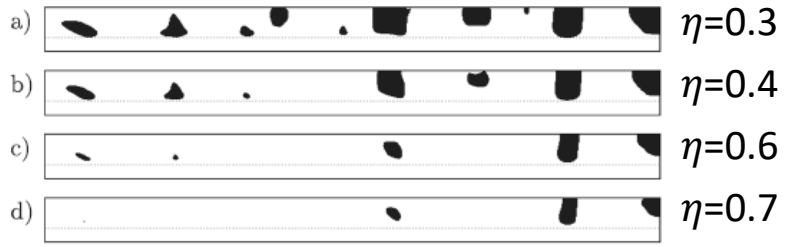
(i) Smoothed design variables



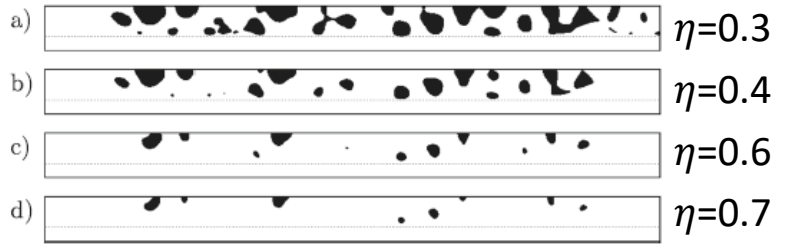
(i) Excitation frequency, $f = 34.36$ Hz.



(ii) Excitation frequency, $f = 51.32$ Hz.



(iii) Excitation frequency, $f = 69.42$ Hz.



(iv) Excitation frequency, $f = 206.3$ Hz.

Fig. 6 Projection of smoothed design variables for the four cases presented in Fig. 5i at different η-values: **a)** $\eta = 0.3$, **b)** $\eta = 0.4$, **c)** $\eta = 0.6$, **d)** $\eta = 0.7$

The double filter approach

- The double filter approach creates geometries which are more likely to have the same topology than the ones in the robust approach.
- This is done by using more filtering and projection operations.
- These operations are computationally cheap, so it does not add much to the computation time of a single iteration, compared to the robust approach.
- (a) shows the design variables and (b) the filtered variables, using a filter radius r_1.
- The filtered variables are projected with strength β_1 and threshold η_1 to obtain (c).
- "Eroded" and "dilated" design realizations are generated by filtering and projecting the design in (c) using a filter radius r_2, strength β_2, and several different thresholds, η_2, from a set η, with $\eta_2 \in \eta$.
- Figures (e) and (f) show this for two thresholds, $\eta_2 \in \{0.3, 0.7\}$.
- $r_2 = \frac{1}{2} r_1$, $\beta_2 = \frac{1}{2}\beta_1$ and $\eta_1 = \min \eta$ is used in the paper.
- A "nominal" design can be obtained using $\eta_2 = 0.5$, which should give a design "in between" (e) and (f).
- The nominal design can also be included in the set of η_2 values but need not be. If it is not, it can be generated at the very end of the optimization.
- Any number of design realizations can be used, but the paper suggest to limit η_2 to the range $0.3 \leq \eta_2 \leq 0.7$.

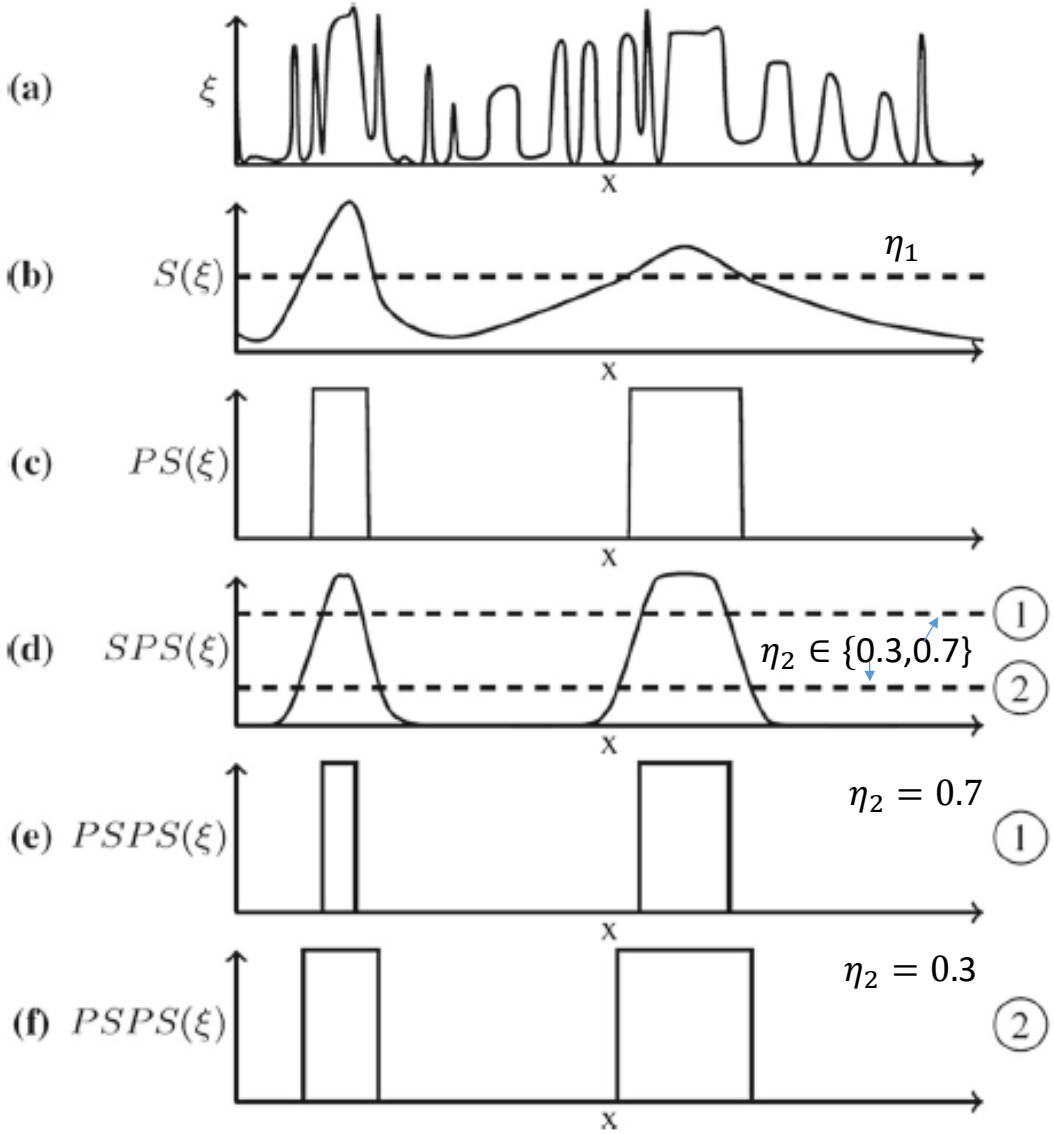
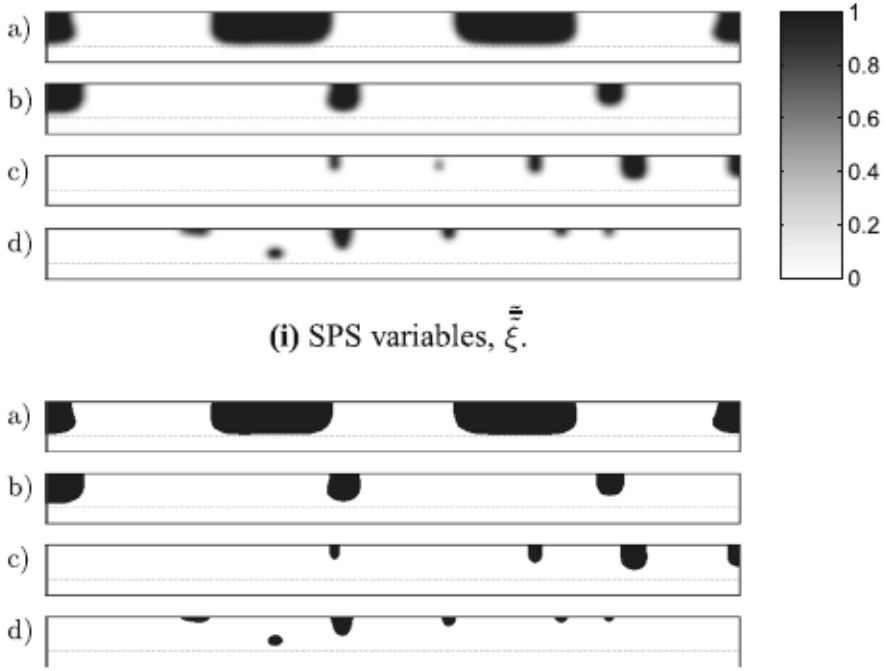


Fig. 7 Sketch of the double filter applied to a function, $\xi(x)$ in 1D (with $\beta_1 \gg 1$, $\beta_2 \gg 1$). Two different projection levels are used for the second projection. $S(\cdot)$ denotes smoothing and $P(\cdot)$ denotes projection. **a** $\xi(x)$, **b** $S_1(\xi(x))$, **c** $P_1(S_1(\xi(x)))$, **d** $S_2(P_1(S_1(\xi(x))))$, **e** $P_{2,1}(S_2(P_1(S_1(\xi(x)))))$, **f** $P_{2,2}(S_2(P_1(S_1(\xi(x)))))$

The double filter standard approach

- The double filter approach with only one design realization
$$\eta_2 \in \eta \text{ with } \eta = \{0.5\}$$
is named *the double filtered standard approach*.
- It is seen to remove the many intermediate values in the double filtered design variables.



(i) SPS variables, $\bar{\bar{\xi}}$.

(ii) Final design (PSPS variables projection at $\eta_2 = 0.5$).

Fig. 8 **(i)** SPS and **(ii)** PSPS design variables for the four excitation frequencies, **a)** $f = 34.36$ Hz, **b)** $f = 51.32$ Hz, **c)** $f = 69.42$ Hz, **d)** $f = 206.3$ Hz. The standard approach with the double filter has been used for the optimization

(S denotes smoothing and P projection)

The double filter approach

The double filter approach is seen to create consistent topologies for different projection thresholds, thus fixing the shortcoming of the standard robust approach.

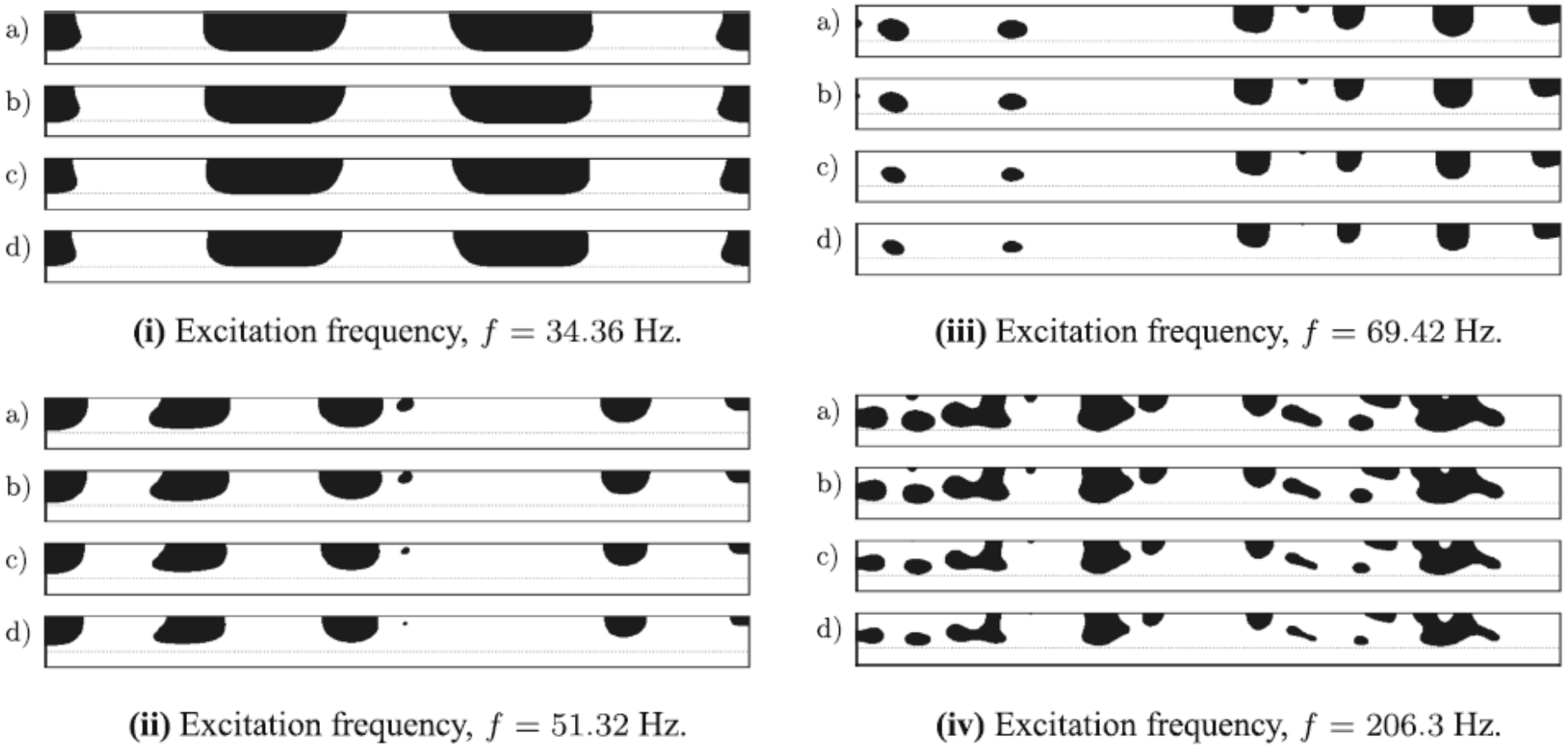


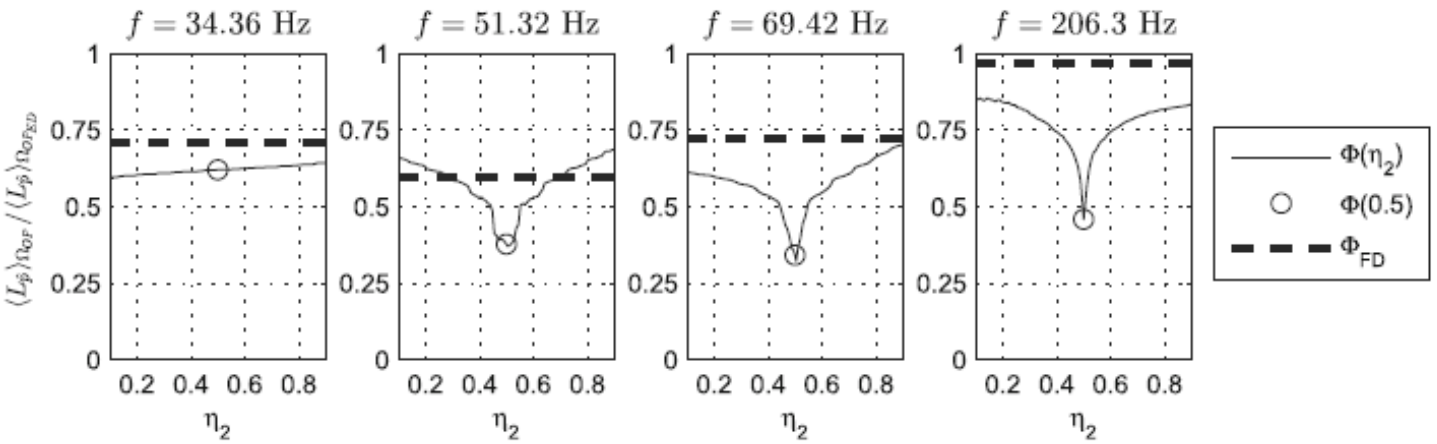
Fig. 11 Projection of SPS design variables at four different η_2-values, **a)** $\eta_2 = 0.3$, **b)** $\eta_2 = 0.4$, **c)** $\eta_2 = 0.6$, **d)** $\eta_2 = 0.7$ for the four designs shown in Fig. 10i

The double filter approach

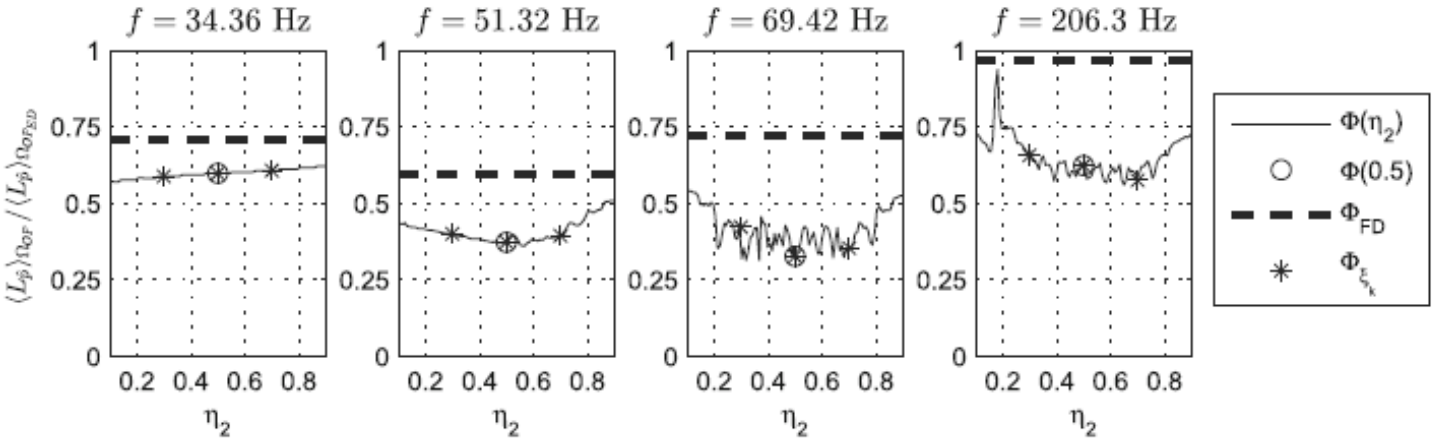
- After the optimization, the designs from the double filter approach and the double filter standard approach were evaluated for different projection thresholds, to create near uniform geometric variations with the same overall topology.
- The results show that the double filter approach yields designs less sensitive to variations than the double filter standard approach.
- The paper also considers a penalization approach to get the similar results, but the method requires careful tuning of a penalty parameter.
  - A standard trick to reduce 'grey' regions is to add $\alpha \int \tilde{\rho}(1-\tilde{\rho}) d\Omega$ to the cost function.
  - α is a penalty factor.
  - This adds a penalty if ρ is not 0 or 1.
  - ρ can also be used instead of $\tilde{\rho}$ in the penalty term.

Fig. 9 $\langle L_{\hat{p}} \rangle_{\Omega_{OP}} / \langle L_{\hat{p}} \rangle_{\Omega_{OP_{ED}}}$ as a function of projection level. **(i)** and **(ii)**: $\langle L_{\hat{p}} \rangle_{\Omega_{OP}}(\eta_2) / \langle L_{\hat{p}} \rangle_{\Omega_{OP_{ED}}}$ is denoted by the thin black line. The performance of the nominal design ($\eta_2 = 0.5$) is denoted by ○. For comparison, $\langle L_{\hat{p}} \rangle_{\Omega_{OP_{FD}}} / \langle L_{\hat{p}} \rangle_{\Omega_{OP_{ED}}}$ is denoted by a thick dashed black line. **(ii)**: The realisations of η_2 for which the design has been optimized are denoted with *

OP: Optimized
ED: Empty Domain
FD: Full Domain



(i) Performance for designs presented in figure 8 obtained using the double filtered standard approach.



(ii) Performance for designs presented in figure 10 obtained using the double filtered robust approach.

Non-uniform design variations

- The double filter approach works well for creating designs which are robust under near uniform design changes (erosion and dilation).
- The paper also considers non-uniform design variations, by making η a function of the spatial coordinates:

$$\eta_2(\mathbf{x}) = \eta_{\min} + (\eta_{\max} - \eta_{\min}) \cdot \mathrm{P}(A \cdot \cos(Bx + C)). \quad (31)$$

  for different values of A, B and C.
- This gives many realizations in the optimization problem, and a more efficient numerical method is used, a hybrid FEM and Wave Based method.
- This method allows for a very coarse grid outside the design domain.
- The used software only allows the Neumann-conditions (the sound source) to be on the boundary of the domain, so the geometry is modified slightly.
- Placing the Neumann-conditions on the boundary of the domain is also easiest to use in FEniCS.
- We will not go further into details with the non-uniform variations, except note that the double filter approach also works in this case.

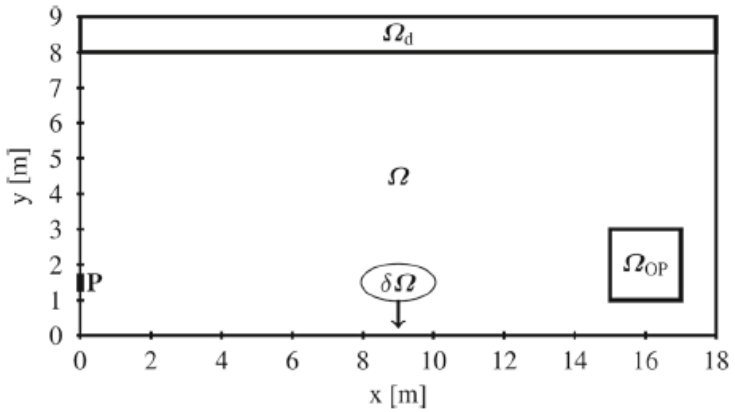


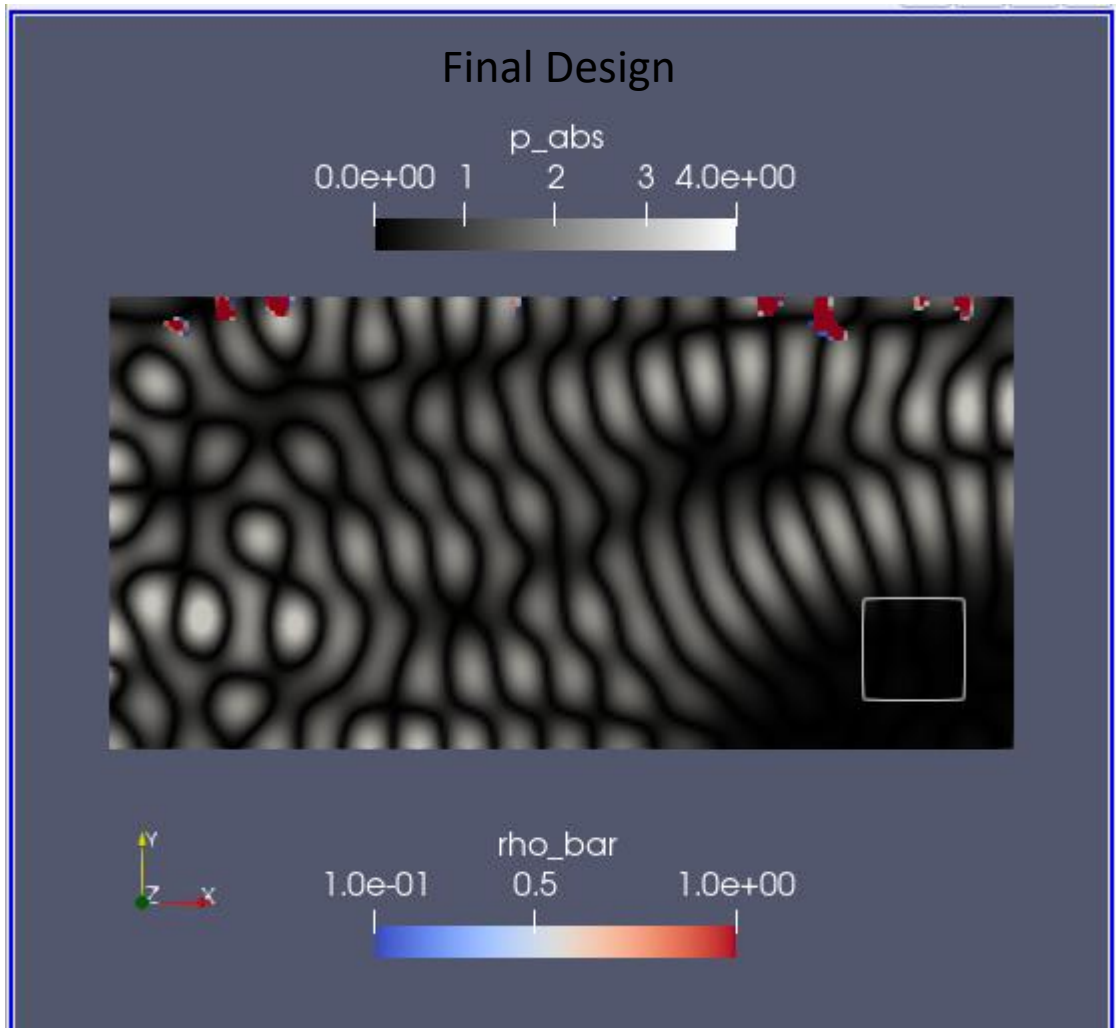
Fig. 14 Illustration of the modified model problem domain. $\Omega_{OP} \in [15.5, 16.5] \times [1.5, 2.5]$ is the optimization domain, $\Omega_d \in [0, 18] \times [8, 9]$ is the design domain and $\mathbf{P} \in 0 \times [1.3, 1.7]$ denotes the region where an acoustic excitation is imposed

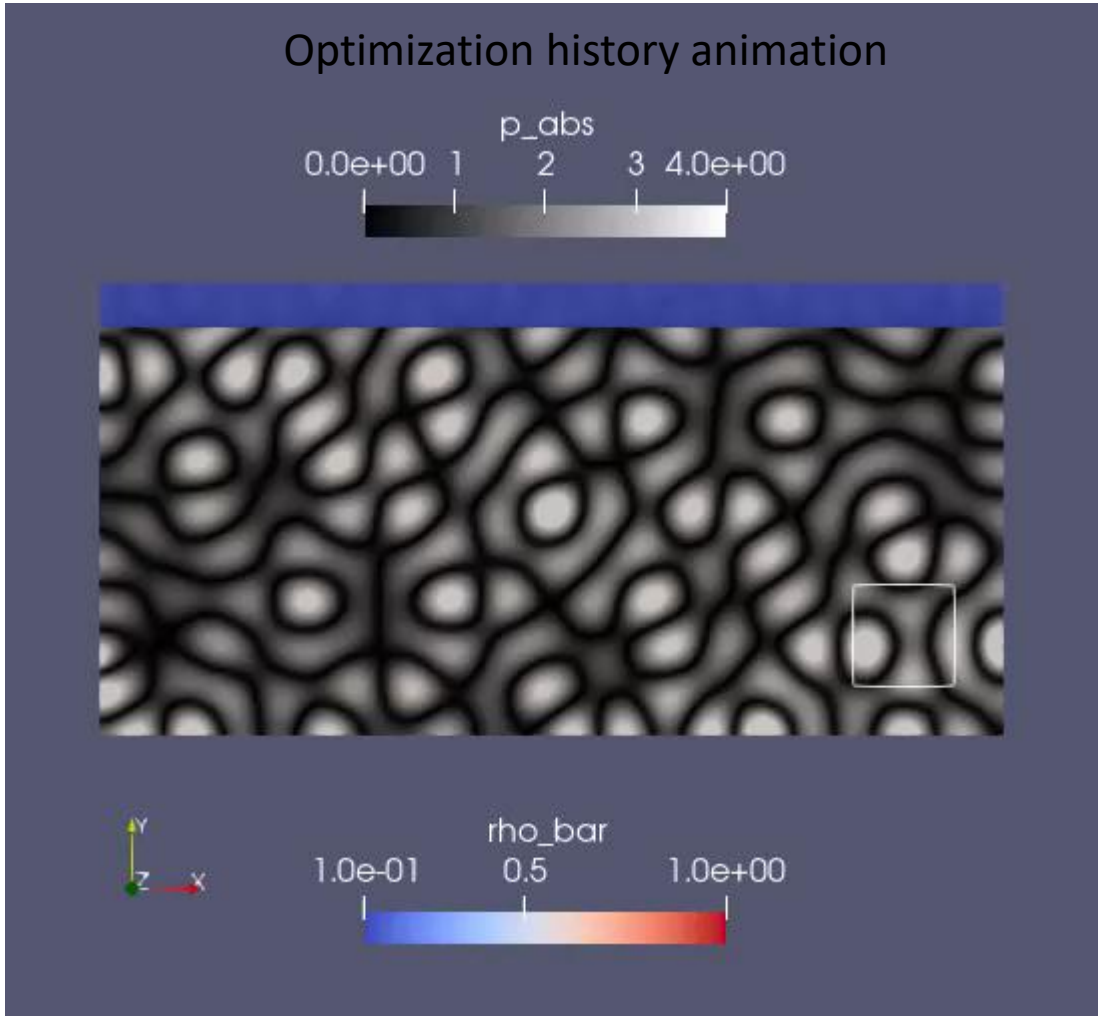
Conclusion

- The topology optimization framework can also be used for wave propagation problems.
- The problems may be more problematic than the standard linear mechanics problems.
- Different material interpolation schemes are often used.
- The original robust topology optimization approach has some deficiencies with wave propagation problems, as they gave filtered variables close to the nominal projection threshold and different η–values do not lead to the same overall topology.
- A double filter approach was developed, to give near uniform design variations, even for filtered variables close to the projection threshold.
- More filtering and projection operations are used, compared to the standard robust approach.
- Most often, the filtering operations are computationally cheap compared to the physics problem considered, and the added complexity usually does not lead to a huge increase in the computation time of an iteration, compared to the standard robust approach.
- The double filter approach can also be used with all the standard problems, like MBB, compliant mechanisms etc.

Some sample calculations from the FEniCS world

Non-robust optimization in our setup, f=206.3 Hz



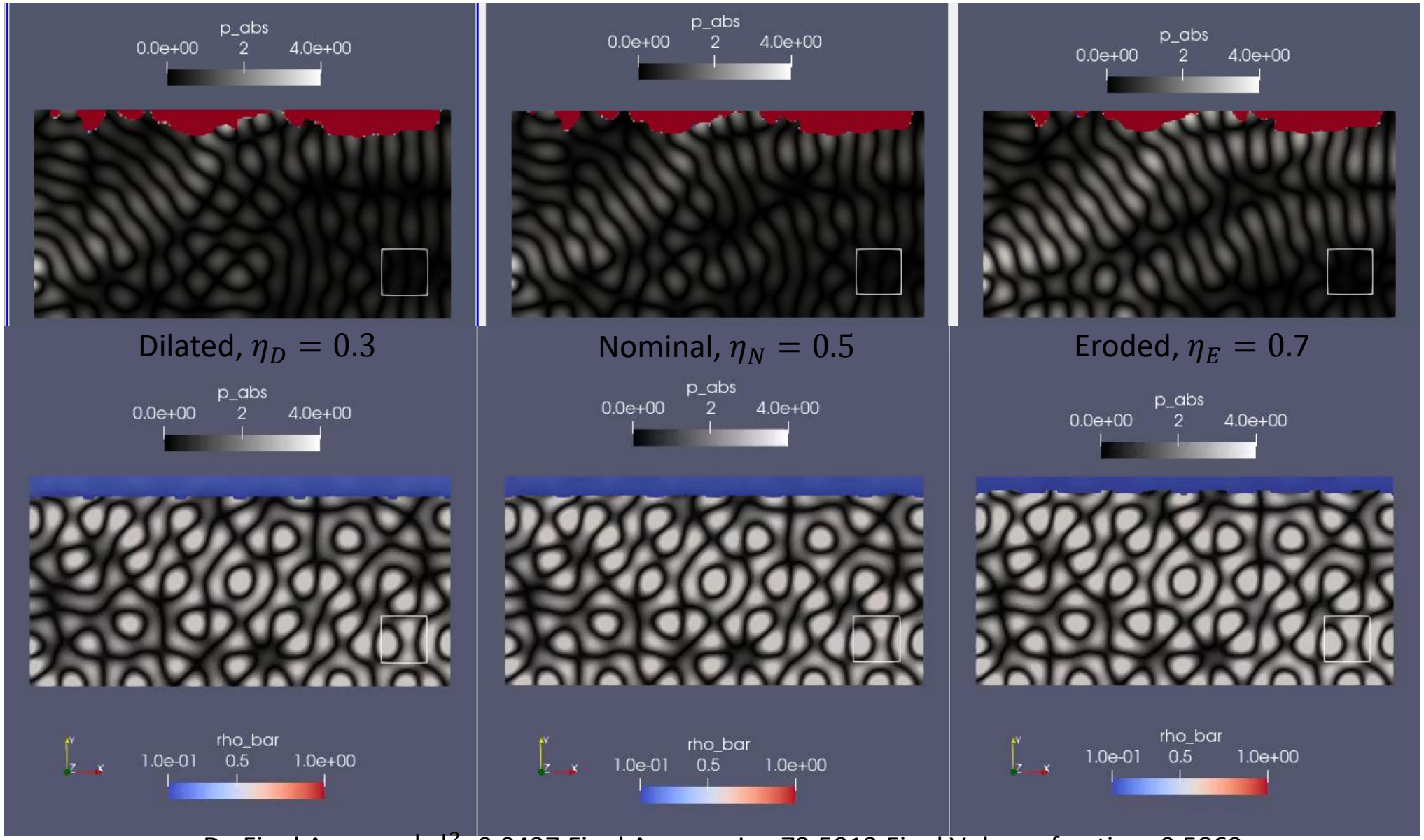


Final Average $|p|^2$: 0.0097 Final Average Lp: 64.0195 Final Volume fraction: 0.0673

Double Filtered Robust optimization in our setup, f=206.3 Hz

Final Design

Optimization history animation

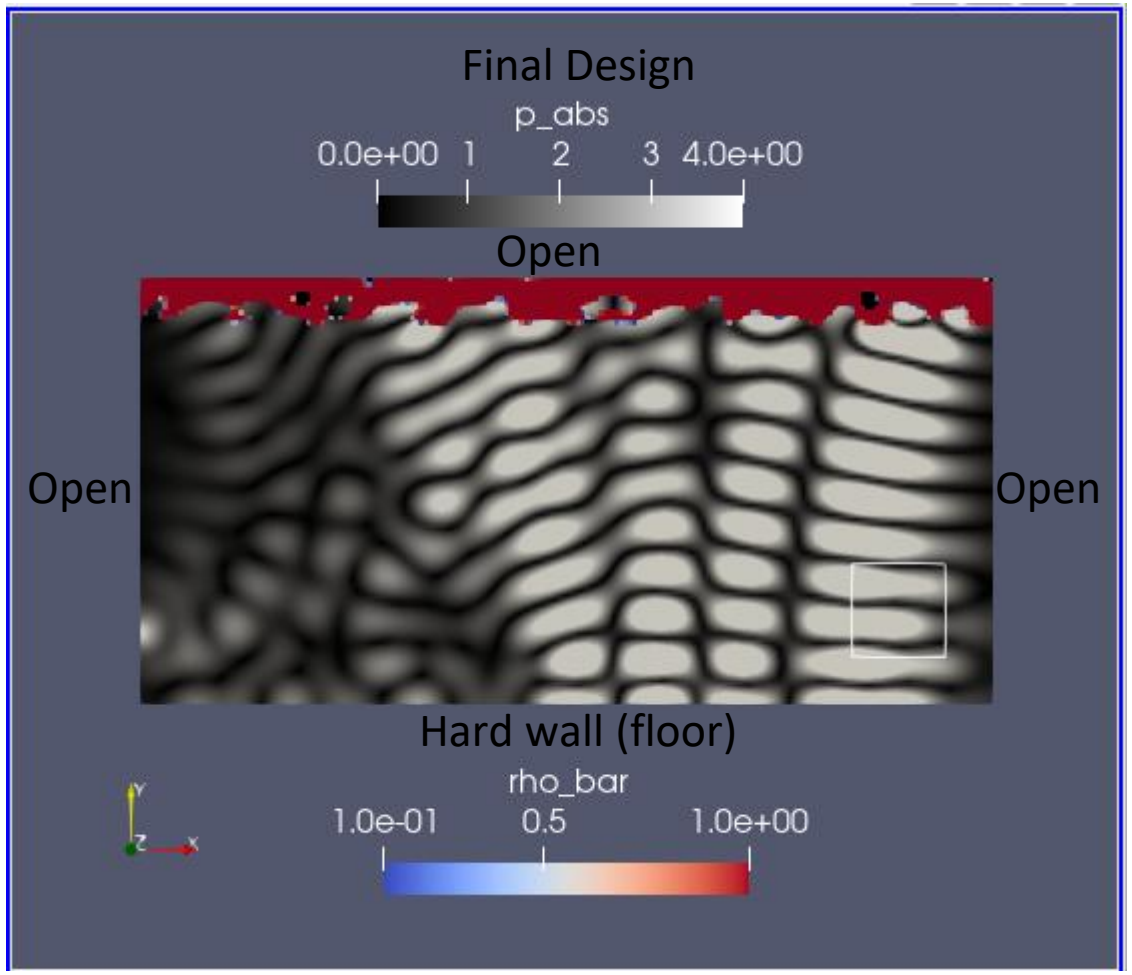


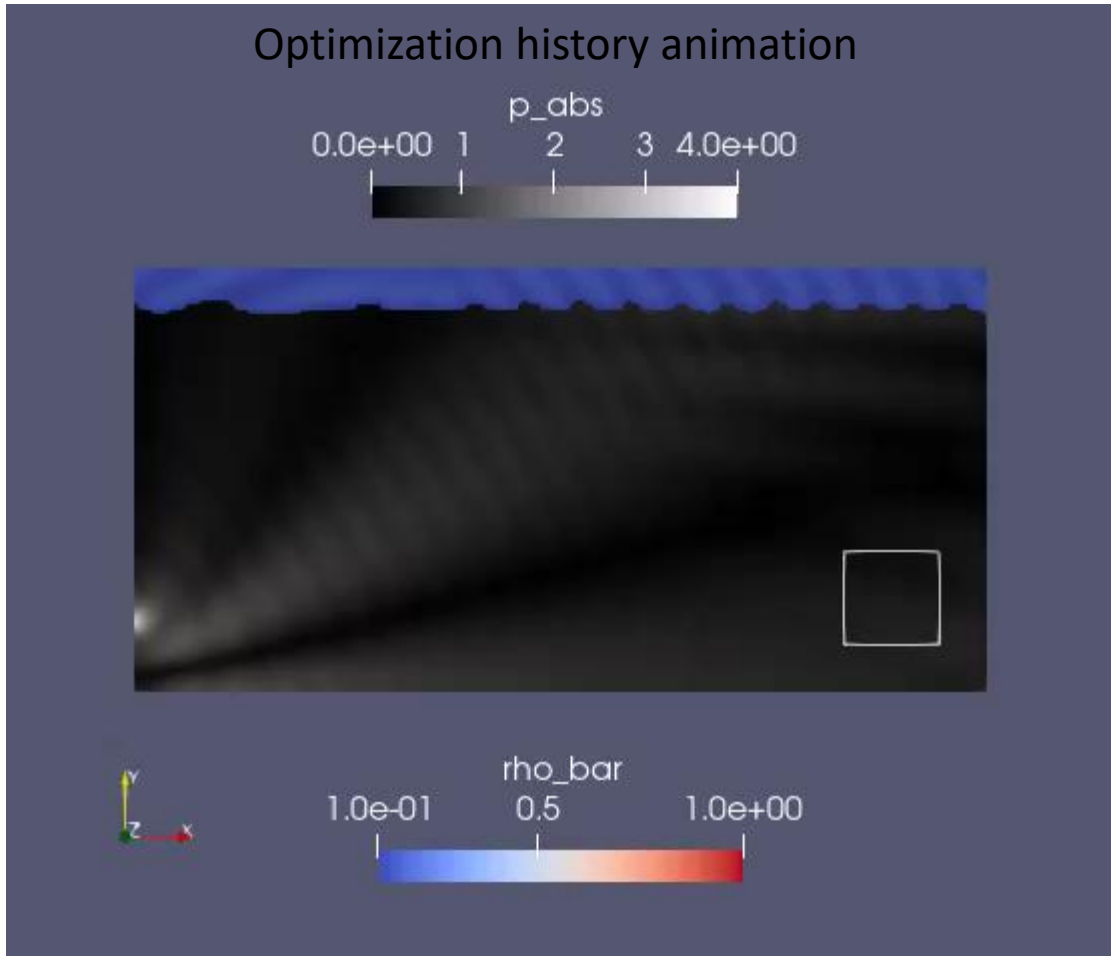
D : Final Average $|p|^2$: 0.0427 Final Average Lp: 73.5013 Final Volume fraction: 0.5869

N : Final Average $|p|^2$: 0.0425 Final Average Lp: 72.9934 Final Volume fraction: 0.5031

E : Final Average $|p|^2$: 0.0427 Final Average Lp: 71.5743 Final Volume fraction: 0.4101

Non-robust maximization of the pressure with open boundaries, f=206.3 Hz



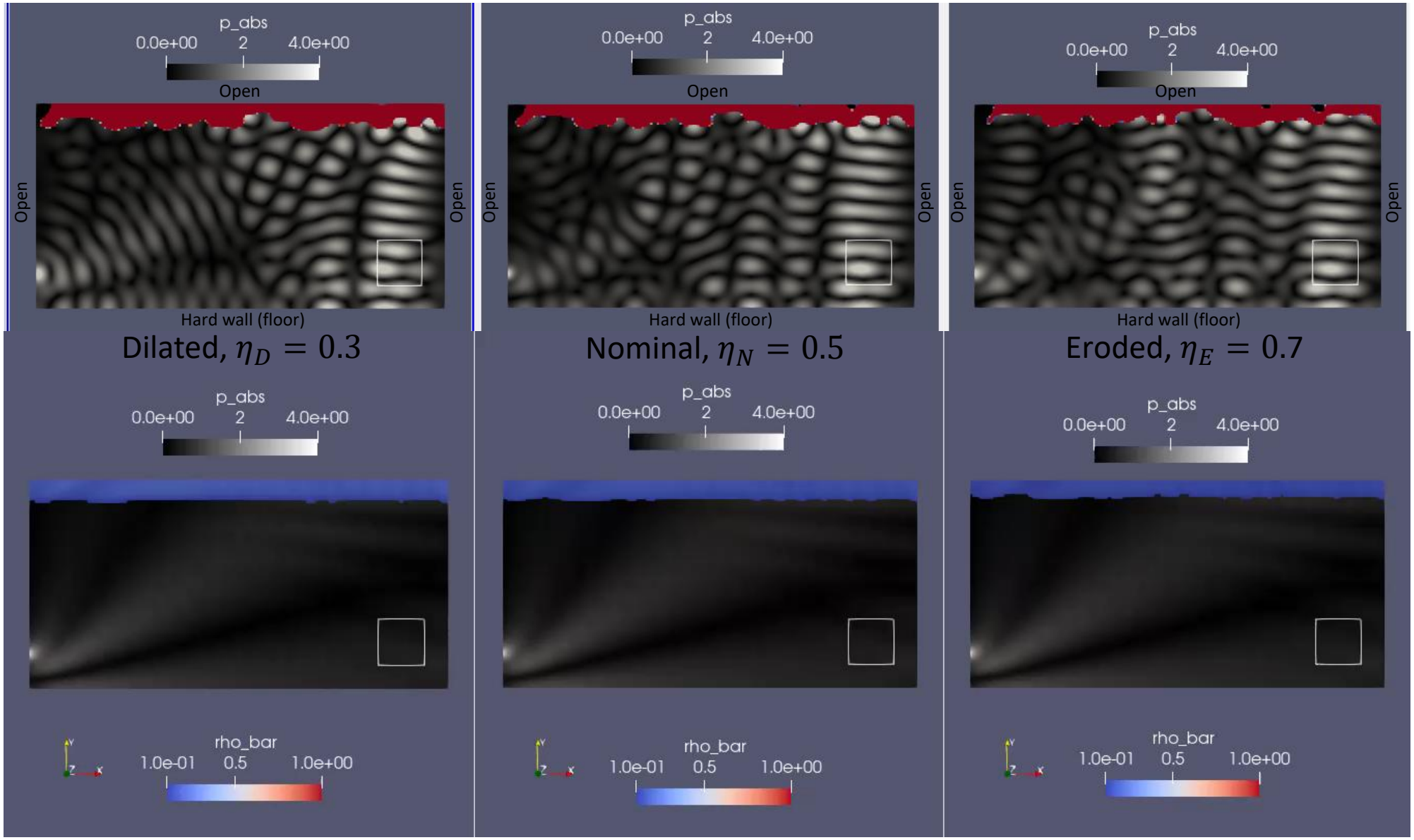


Final Average $|p|^2$: 15.9327 Final Average Lp: 99.3702 Final Volume fraction: 0.7324

Double Filtered Robust maximization of the pressure with open boundaries, f=206.3 Hz

Final Design

Optimization history animation

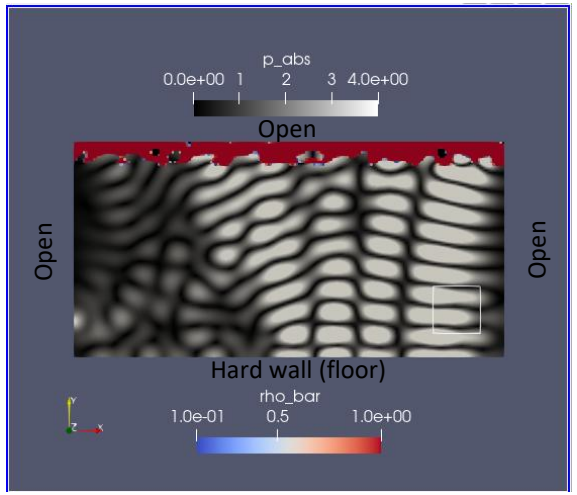


D : Final Average $|p|^2$: 5.4416 Final Average Lp: 95.5891 Final Volume fraction: 0.8439
N : Final Average $|p|^2$: 5.4417 Final Average Lp: 95.6351 Final Volume fraction: 0.7511
E : Final Average $|p|^2$: 5.4418 Final Average Lp: 95.7757 Final Volume fraction: 0.6338

Non-robust vs. Double Filtered Robust Maximum Pressure

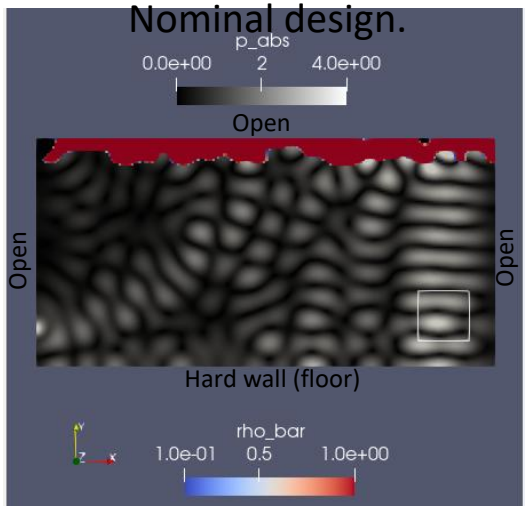
- Surprisingly, the non-robust has a lower average pressure than the robust with very small $\Delta\eta = \eta_E - \eta_D$.
- The non-robust has thus converged to a worse local optimum than the robust optimization.

Non-robust



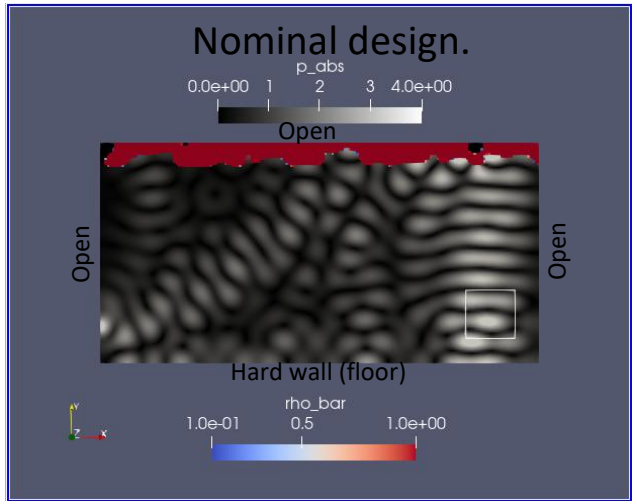
Final Average $|p|^2$: 15.9327

Robust: $\Delta\eta = 0.4$



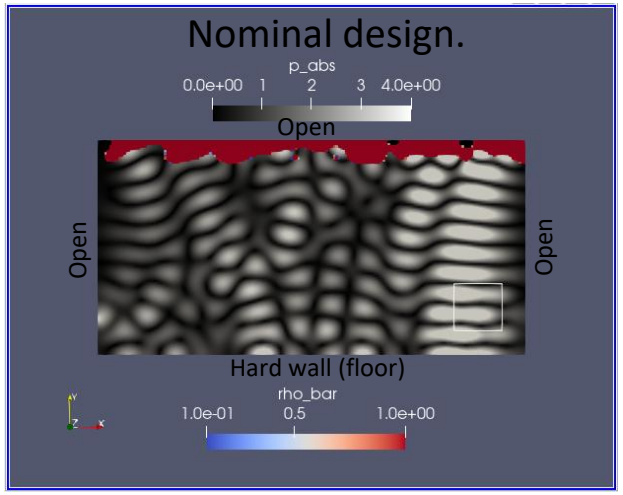
D : Final Average $|p|^2$: 5.4416
N : Final Average $|p|^2$: 5.4417
E : Final Average $|p|^2$: 5.4418

Robust: $\Delta\eta = 0.2$



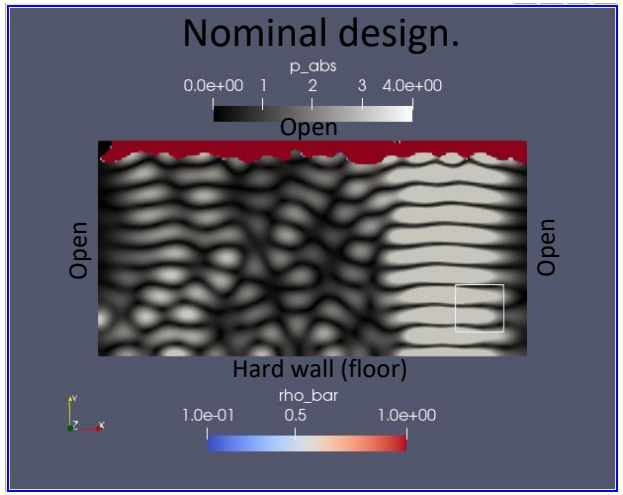
N : Final Average $|p|^2$: 5.6669
D : Final Average $|p|^2$: 5.6666
E : Final Average $|p|^2$: 5.6667

Robust: $\Delta\eta = 0.1$



N : Final Average $|p|^2$: 11.0100
D : Final Average $|p|^2$: 11.0123
E : Final Average $|p|^2$: 11.0107

Robust: $\Delta\eta = 0.02$



N : Final Average $|p|^2$: 20.2436
D : Final Average $|p|^2$: 20.1635
E : Final Average $|p|^2$: 20.1590

Implementation of the Time-Harmonic Complex Wave Equation

The time-harmonic wave equation

- The equation considered in the paper is $\nabla \cdot (\rho^{-1}\nabla p) + \omega^2 K^{-1} p = 0$ with velocity equation $i\omega \boldsymbol{u} + \rho^{-1}\nabla p = 0$.
- $p = p_r + ip_i$ is the complex time-harmonic pressure amplitude.
- The version of FEniCS we are using does not support complex numbers, so we must convert our expressions to real expressions. 'FEniCSx' support complex numbers but dolfin-adjoint has not yet been made for this, so that does not help us much. Unless you want to make your own adjoint code… ☺
- We assume ρ is real but $K = K_r + iK_i$ may be complex. Since K^{-1} enters the equation, we use $|K|^2 = KK^*$ to write

$$K^{-1} = \frac{1}{K} = \frac{K^*}{|K|^2} = \frac{K_r - iK_i}{K_r^2 + K_i^2} = K_r^{-1} + iK_i^{-1} \Rightarrow \quad K_r^{-1} = \frac{K_r}{K_r^2 + K_i^2} \ , \qquad K_i^{-1} = \frac{-K_i}{K_r^2 + K_i^2}$$

- Since p is complex, the wave equation is in fact two equations, one for the real part and one for the imaginary part.
- The wave equation is split into its real and its imaginary parts:

$$\nabla \cdot (\rho^{-1}\nabla p) + \omega^2 K^{-1} p = \nabla \cdot \left(\rho^{-1}\nabla(p_r + ip_i)\right) + \omega^2\left(K_r^{-1} + iK_i^{-1}\right)(p_r + ip_i)$$
$$= \nabla \cdot (\rho^{-1}\nabla p_r) + \omega^2\left(K_r^{-1}p_r - K_i^{-1}p_i\right) + i\left(\nabla \cdot (\rho^{-1}\nabla p_i) + \omega^2\left(K_r^{-1}p_i + K_i^{-1}p_r\right)\right) = 0 + i0$$

- The two equations are thus (after multiplying with -1)

$$-\nabla \cdot (\rho^{-1}\nabla p_r) - \omega^2\left(K_r^{-1}p_r - K_i^{-1}p_i\right) = 0, \qquad -\nabla \cdot (\rho^{-1}\nabla p_i) - \omega^2\left(K_r^{-1}p_i + K_i^{-1}p_r\right) = 0$$

- These must be solved for p_r and p_i together with the appropriate boundary conditions.

The time-harmonic wave equation

- The weak forms are derived by multiplying each with test functions, t_r and t_i, and integrated (as usual):

$$\int_\Omega \left(-\nabla \cdot (\rho^{-1}\nabla p_r) - \omega^2\left(K_r^{-1}p_r - K_i^{-1}p_i\right)\right) t_r d\Omega = 0,$$

$$\int_\Omega \left(-\nabla \cdot (\rho^{-1}\nabla p_i) - \omega^2\left(K_r^{-1}p_i + K_i^{-1}p_r\right)\right) t_i d\Omega = 0$$

- Partial integration is performed to get

$$\int_\Omega \left(\nabla t_t \cdot (\rho^{-1}\nabla p_r) - \omega^2\left(K_r^{-1}p_r - K_i^{-1}p_i\right)t_r\right)d\Omega - \int_{\partial\Omega} \hat{n} \cdot (\rho^{-1}\nabla p_r)t_r \, dS = 0,$$

$$\int_\Omega \left(\nabla t_i \cdot (\rho^{-1}\nabla p_i) - \omega^2\left(K_r^{-1}p_i + K_i^{-1}p_r\right)t_i\right)d\Omega - \int_{\partial\Omega} \hat{n} \cdot (\rho^{-1}\nabla p_i)t_i \, dS = 0$$

- The Neumann terms can be found from the velocity equation

$$i\omega\boldsymbol{u} + \rho^{-1}\nabla p = 0 \;\; \Rightarrow \;\; \rho^{-1}\nabla p = -i\omega\boldsymbol{u} \;\Rightarrow \rho^{-1}\nabla(p_r + ip_i) = -i\omega(\boldsymbol{u}_r + i\boldsymbol{u}_i) \;\Rightarrow$$
$$\rho^{-1}\nabla p_r = \omega\boldsymbol{u}_i \;, \qquad \rho^{-1}\nabla p_i = -\omega\boldsymbol{u}_r.$$

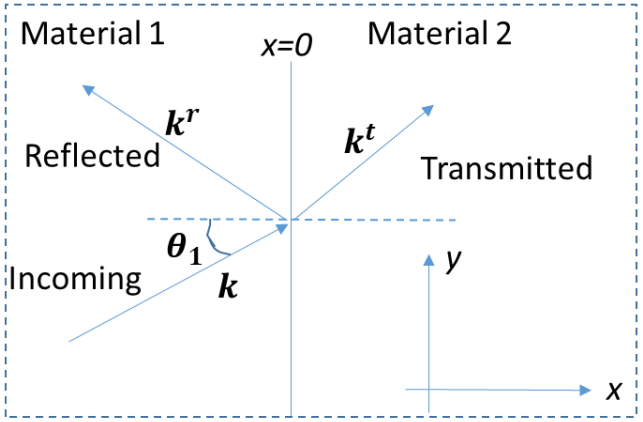
  for a prescribed (complex) velocity amplitude $\boldsymbol{u}$.

- If $\boldsymbol{u}$ is non-zero, the boundary is vibrating and generating pressure waves, like a loudspeaker.

Boundary Conditions

- Remember, the reflection factor for a normal incidence plane wave

$$r(\theta_1 = 0) = \frac{R}{I} = \frac{Z_2 - Z_1}{Z_2 + Z_1}, \qquad Z_1 = \rho_1 c_1 \text{ and } Z_2 = \rho_2 c_2$$



- At a hard wall $Z_2 \gg Z_1$. The velocity is zero and the Neumann term is zero. This makes the wave reflect from the wall with $r = 1$.
- At a soft wall $Z_1 \gg Z_2$ and $p = 0$ (Dirichlet boundary condition). This also reflects the wave, but with $r = -1$.
- To mimic the case of no wall, an open domain can be used (also named scattering boundary condition or absorbing boundary condition or radiation boundary condition).

Computational Domain

- Plane wave radiation boundary conditions makes the reflection of a normal incident plane wave zero. A first order absorbing boundary the condition is

$$\hat{n} \cdot (\rho^{-1} \nabla p) = -ik\rho^{-1} p = -i \frac{\omega \rho^{-1}}{c} p \;\Rightarrow$$

$$\hat{n} \cdot (\rho^{-1} \nabla p_r) = \omega \rho^{-1} \left(c_r^{-1} p_i + c_i^{-1} p_r \right), \qquad \hat{n} \cdot (\rho^{-1} \nabla p_i) = -\omega \rho^{-1} (c_r^{-1} p_r + c_i^{-1} p_i)$$

Computational Domain

- Remember that $K = \rho c^2$.
- Alternative "Perfectly Matched Layers" (PML) can be used, which add a layer of artificial material with impedance Z_2, which absorbs the wave but does not create reflections at the $Z_1 \rightarrow Z_2$ interface. We will not use that, as it is more complicated.

The source is placed on a small part of the boundary and given by $\boldsymbol{u}_r = \boldsymbol{u}_{\text{source}}, \boldsymbol{u}_i = \boldsymbol{0}$. It radiates in "all" directions.

Hard wall reflecting boundary conditions

$$\hat{n} \cdot (\rho^{-1} \nabla p_r) = 0$$
$$\hat{n} \cdot (\rho^{-1} \nabla p_i) = 0$$

A lot of reflections are seen, resulting in standing waves in the cavity.

Note the many variations in the sound pressure. The mesh needs to be fine enough to make the function space provide an accurate description of this complicated function. Often the mesh size is set as a fraction of the wavelength.

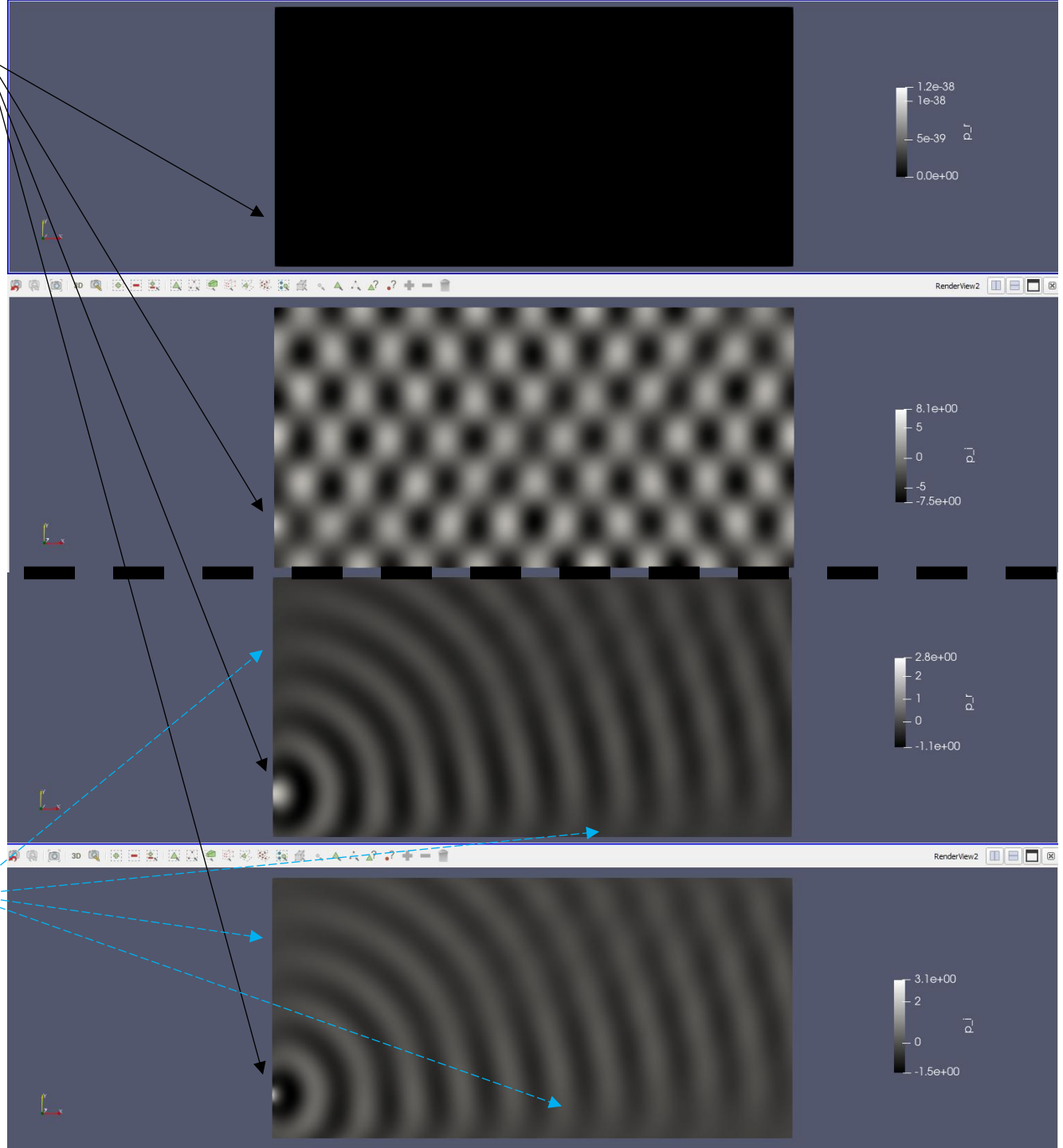
Open boundary conditions

$$\hat{n} \cdot (\rho^{-1} \nabla p_r) = \omega \rho^{-1} \left(c_r^{-1} p_i + c_i^{-1} p_r \right)$$
$$\hat{n} \cdot (\rho^{-1} \nabla p_i) = -\omega \rho^{-1} (c_r^{-1} p_r + c_i^{-1} p_i)$$

The source radiates in "all" directions and is largely undisturbed by the boundary

Some small "spurious" reflections are seen from the left and bottom boundaries.

These are expected, since the radiation open boundary condition only works perfectly for a wave at normal incidence.



$p_r \approx 0$

p_i

p_r

p_i

Integration domains

```
Wx=18
Wy=9
mesh_size=0.05
self.mesh = RectangleMesh.create([Point(0, 0), Point(Wx, Wy)], [int(Wx/mesh_size), int(Wy/mesh_size)], cell_type)
epsilon=1e-6

# Exterior excitation boundary
size_ext=0.4
class input_boundary_exterior_2D(SubDomain):
    def inside(self, x, on_boundary):
        return on_boundary and near(x[0],0) and (x[1]>=1.3-epsilon) and (x[1]<=1.3+size_ext+epsilon)

# Target domain
class target_2D(SubDomain):
    def inside(self, x, on_boundary):
        return x[0]>=15-epsilon and x[0]<=17+epsilon and x[1]>=1-epsilon and x[1]<=3+epsilon

self.boundary_parts = MeshFunction("size_t", self.mesh, 1)
self.boundary_parts.set_all(0)
self.domain_parts = MeshFunction("size_t", self.mesh, 2)
self.domain_parts.set_all(0)

target_2D().mark(self.domain_parts,1)
input_boundary_exterior_2D().mark(self.boundary_parts,2)

# Define new surface measure with marked surfaces
self.dx = Measure('dx', domain=self.mesh, subdomain_data=self.domain_parts)
self.ds = Measure('ds', domain=self.mesh, subdomain_data=self.boundary_parts)

# Write boundary and domain parts for inspection in Paraview
File(self.directory+'boundary_parts.pvd') << self.boundary_parts
File(self.directory+'domain_parts.pvd') << self.domain_parts
```

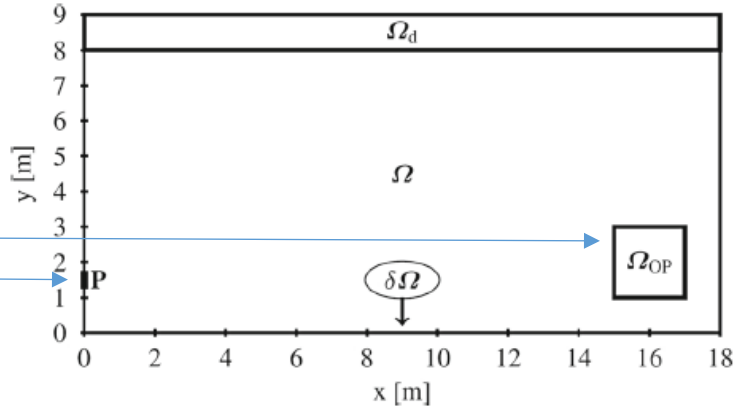


Fig. 14 Illustration of the modified model problem domain. $\Omega_{OP} \in [15.5, 16.5] \times [1.5, 2.5]$ is the optimization domain, $\Omega_d \in [0, 18] \times [8, 9]$ is the design domain and $\mathbf{P} \in 0 \times [1.3, 1.7]$ denotes the region where an acoustic excitation is imposed

Cost function example:

```
self.J=self.f_scale*(p_r**2+p_i**2)/self.TargetVol*self.mech_group.dx(1)
```

Implementation of the weak form

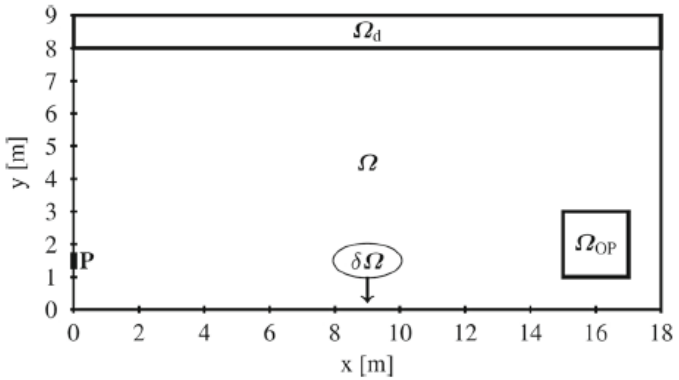
$$\int_\Omega \left(\nabla t_t \cdot (\rho^{-1}\nabla p_r) - \omega^2\left(K_r^{-1}p_r - K_i^{-1}p_i\right)t_r\right)d\Omega - \int_{\partial\Omega} \hat{n}\cdot(\rho^{-1}\nabla p_r)t_r\, dS = 0, \qquad \int_\Omega \left(\nabla t_i \cdot (\rho^{-1}\nabla p_i) - \omega^2\left(K_r^{-1}p_i + K_i^{-1}p_r\right)t_i\right)d\Omega - \int_{\partial\Omega} \hat{n}\cdot(\rho^{-1}\nabla p_i)t_i\, dS = 0$$

$$\rho^{-1}\nabla p = -i\omega\boldsymbol{u} \;\Rightarrow\; \rho^{-1}\nabla p_r = \omega\boldsymbol{u}_i\;, \rho^{-1}\nabla p_i = -\omega\boldsymbol{u}_r$$

```
    # Mixed function space for real and imaginary parts
    self.FE=FiniteElement("P", self.mesh.ufl_cell(), 1)
    self.W=FunctionSpace(self.mesh,self.FE*self.FE) # Mixed Space consisting of two P1 functions
    V_r,V_i=self.W.split()
    self.V_r=V_r.collapse()                         # Function space for p_r
    self.V_i=V_i.collapse()                         # Function space for p_i


def solve(self):
    (t_r,t_i)=TestFunctions(self.W)
    (p_r,p_i)=split( TrialFunction(self.W) )

    f_r=self.omega*np.real(self.U)    # U is the (complex) velocity (input) amplitude normal to the boundary.
    f_i=-self.omega*np.imag(self.U)
    F = (self.inv_rho*inner(grad(p_r),grad(t_r))-self.omega**2*(self.inv_K_r*p_r-self.inv_K_i*p_i)*t_r)*self.dx
    F+= (self.inv_rho*inner(grad(p_i),grad(t_i))-self.omega**2*(self.inv_K_r*p_i+self.inv_K_i*p_r)*t_i)*self.dx
    F+= -t_r*f_i*self.ds(2)-t_i*f_r*self.ds(2)

    a,L=lhs(F),rhs(F)

    # Compute solution
    solution=Function(self.W)
    solve(a == L, solution, self.dbc, solver_parameters={"linear_solver": "mumps"})
    (p_r,p_i)=solution.split(deepcopy=True)
    p_r.rename('p_r','p_r')
    p_i.rename('p_i','p_i')
    return (p_r,p_i)
```

Optimizing only on the top part of the domain

```
def InsideDesignDomain(self,x):
    # Return true if coordinates in x is inside design domain
    return x[1]>=8

def getDDmask(self,rho,Vd):
    # Creates mask which is 1 inside the design domain and 0 everywhere else.
    # rho is a function from Vd.
    #
    # Get rho's DOF coordinates
    coor = Vd.tabulate_dof_coordinates().reshape((-1, self.dim))
    size=rho.vector().get_local().shape[0]
    i=0
    DD=np.zeros(size)
    regions=np.zeros(size)
    for c in coor:
        if self.InsideDesignDomain(c):
            DD[i]=1.0
        else:
            DD[i]=0.0
        i=i+1
    return DD
```



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```
def setUpOptimizer(self,x0):
    """Initializes MMA"""
    self.mma=MMA_petsc(x0,3+1,f=self.f,g=self.g,plot_k=self.plot_k)

    self.mma.xmin[:]=0.0
    self.mma.xmax[:]=self.DDmask

    # MMA setup
    self.mma.setMinMax(3,1) # Use min-max for robust version
    self.mma.move=0.1
    self.mma.xtol=0.01
    self.mma.ftol=0.001
    self.mma.lmax=5
    self.mma.kmax=60 # More iterations than usual!
    self.mma.kmin=10
    self.mma.kav=10
    self.mma.it=0
    self.mma.mma_timing=False
```